



AI SHORT FILM PRODUCTION

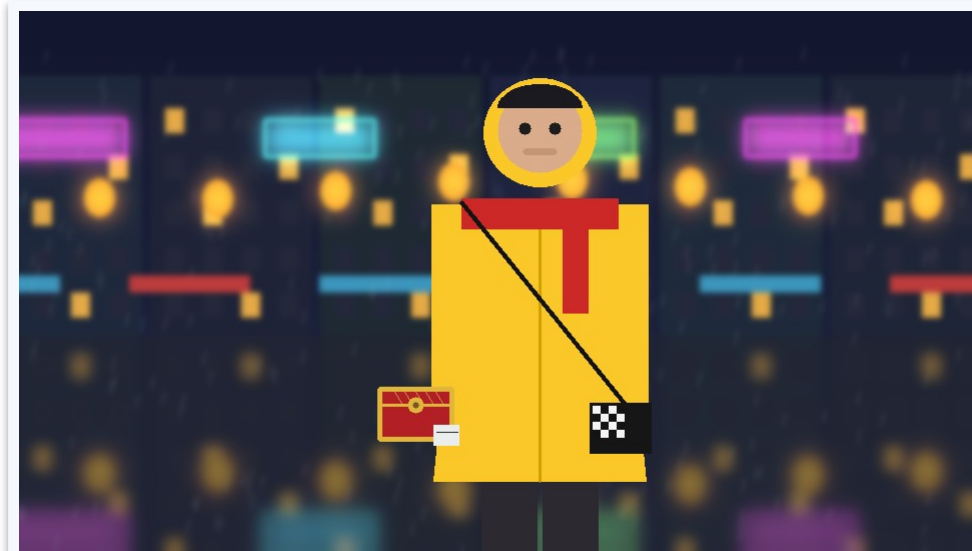
COURSE SLIDES · WSQ

Create Short Video Film using AI

WSQ Course Code: TGS-2020505925 · 2 Days · 16 Programme Hours
Skills Framework TSC: Computer Vision Technology (ICT-DIT-4022-1.1)
Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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The Last Ferry — course film project (SIMULATED illustration)



COURSE ADMINISTRATION

Welcome and Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator displays the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance on the SSG record is required to be eligible for assessment and for funding.
- Attendance is taken three times: AM, PM and again immediately before the assessment block on Day 2.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the
trainer delivering this run

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING AND INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Curriculum Developer and
Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Curriculum Developer and Subject Matter Expert for this course (Course Proposal CA-WSQ-2020-013290).

BACKGROUND

PhD; two decades of engineering, data science and AI practice across industry and training.

DELIVERS

WSQ courses in computer vision, machine learning, deep learning, generative AI and data analytics.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name, organisation and role.
- Any experience you have with video editing, photography, storytelling or AI image and video tools.
- Any experience you have with Python (the labs use it, but every script is supplied and explained).
- One short film or video you would like to be able to plan, generate and finish with AI by the end of Day 2.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

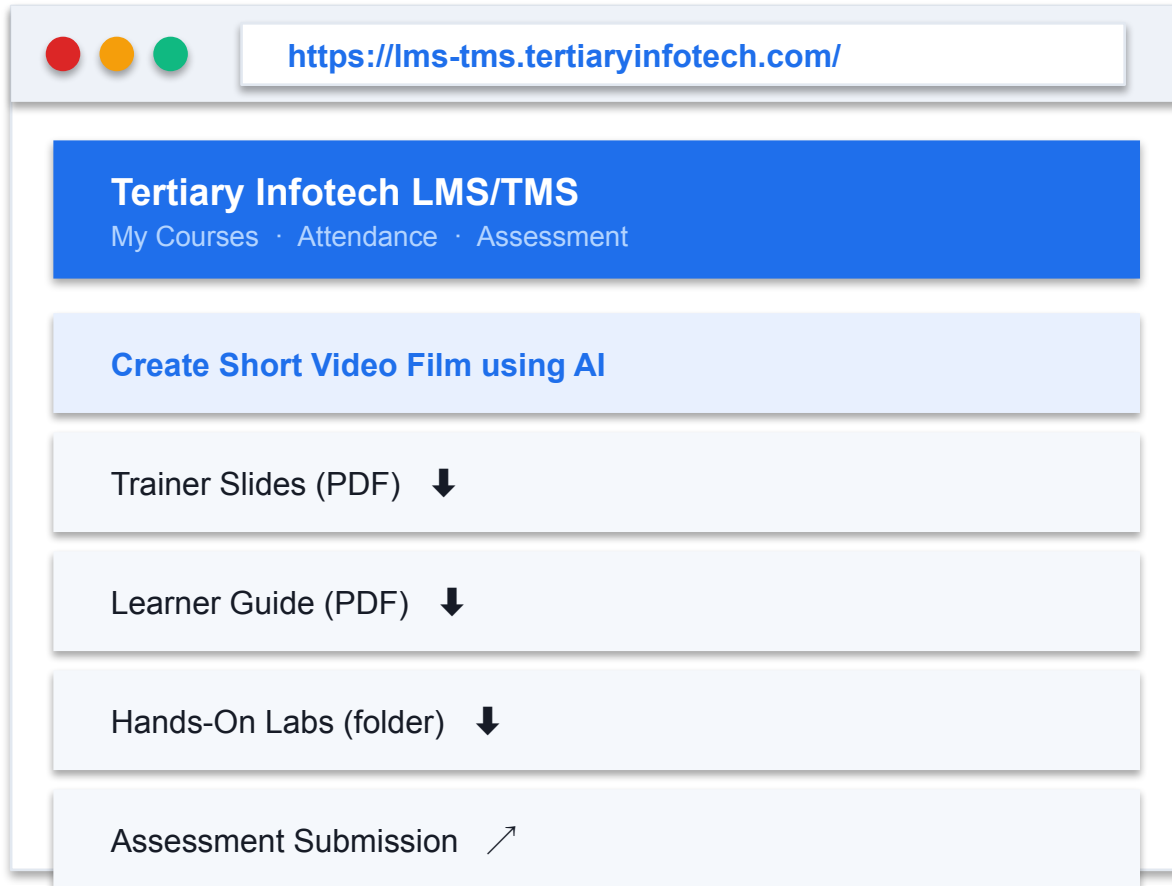
5

Be punctual; return from breaks on time.

6

75% attendance is required for funding eligibility.

Download Course Material



- 1 Sign in**
lms-tms.tertiaryinfotech.com — log in with your registered email (OTP or password).
- 2 Open your course**
Select this course under My Courses.
- 3 Download materials**
Slides, Learner Guide and the labs folder — your open-book references for the assessment.
- 4 Submit and survey**
Upload your completed assessment answers here, and complete the TRAQOM survey.

All course material is downloaded from the LMS/TMS portal. Keep it open: both assessment instruments are open book.

Skills Framework Alignment

Element	Detail
TSC Title	Computer Vision Technology
TSC Code	ICT-DIT-4022-1.1
Proficiency Level	Level 4 — set up and deploy video analytics algorithms and perform system performance evaluations
TSC Description	Develop and deploy vision analytics algorithms and spatial sensing and/or reasoning systems
Assessed knowledge	K1 to K13 — assessed in the Written Assessment
Assessed abilities	A1 to A8 — assessed in the Practical Performance

Every topic, lab and assessment item in this course maps to these TSC knowledge and ability statements. Nothing is assessed that is not taught.

Assessed Knowledge — K1 to K13

Code	Knowledge statement	Taught in
K1	Vision system concepts	Topic 01
K2	Business applications of vision systems	Topic 01
K3	Methods to represent image and video data	Topic 02
K4	Image and video processing, filtering and transformation methods	Topic 02
K5	Feature extraction and representation techniques	Topic 03
K6	Local feature descriptions, edge, colour, texture and motion	Topic 03
K7	Global feature descriptions, statistical and geometrical methods	Topic 03
K8	Deep learning concepts	Topic 04
K9	Object segmentation, detection and recognition	Topic 04
K10	Activity tracking, generative models, scene understanding and event discovery	Topic 04
K11	Vision system architecture	Topic 05
K12	Vision communication protocols	Topic 05
K13	Real-world design constraints and solution options	Topic 05

The Written Assessment carries exactly one question per knowledge statement, in this order.

Assessed Abilities — A1 to A8

Code	Ability statement	Practised in
A1	Identify the needs of vision systems technology in industrial applications	Lab 01
A2	Apply the principles of processing, filtering and analysis methods for video data	Lab 02, Lab 03, Lab 04, Lab 05
A3	Analyse global feature descriptions	Lab 07
A4	Design and implement feature extraction and representation methods	Lab 06
A5	Design and apply machine-learning based methods for object detection, object tracking and activity recognition	Lab 08, Lab 09, Lab 11
A6	Design and apply video analytics algorithms for high-level video analytics tasks	Lab 09, Lab 10, Lab 11
A7	Design the architecture of applied vision systems	Lab 12
A8	Design, develop and evaluate edge-based and cloud-based systems	Lab 12

PP Task 1 covers A1; Task 2 covers A2 and A3; Task 3 covers A4; Task 4 covers A5 and A6; Task 5 covers A7 and A8.

BY THE END OF THIS COURSE YOU WILL BE ABLE TO

Learning Outcomes

1

LO1

Understand basic vision systems concepts and applications

2

LO2

Apply image processing

3

LO3

Implement feature extraction

4

LO4

Apply machine learning based computer vision methods

5

LO5

Implement video analytics algorithms

6

LO6

Evaluate edge vs cloud-based computer vision systems

Course Outline — Five Delivery Topics

Day 1

- **Topic 01 — AI Vision and Generative AI for Short Film Creation**
 - LU1 · ELO1 · K1, K2, A1
- **Topic 02 — AI Image Processing, Generation and Visual Enhancement**
 - LU2 · ELO2 · K3, K4, A2
- **Topic 03 — AI Visual Features, Character Consistency and Scene Design**
 - LU3 · ELO3 · K5, K6, K7, A3, A4

Day 2

- **Topic 04 — AI Video Generation, Animation and Video Analytics**
 - LU4 + LU5 · ELO4 + ELO5 · K8, K9, K10, A5, A6
- **Topic 05 — Cloud and Edge AI Workflows for Short Film Production**
 - LU6 · ELO6 · K11, K12, K13, A7, A8
- **12 hands-on labs**
 - One short film — The Last Ferry — built across all twelve labs

Five delivery topics mapped onto the six approved Learning Units. Topic 04 consolidates LU4 and LU5 concept delivery; both units keep their full practical allocation.

Lesson Plan — Day 1

Time	Min	Component	Session
08:45–09:00	15	ADMIN	Course welcome, attendance, introductions and LMS access; outside programme minutes
09:00–10:15	75	CR	Topic1 / LU1: the AI film pipeline, how generators work, business uses; facilitated Lab01 film n
10:15–10:30	15	BREAK	Tea break; excluded
10:30–11:45	75	CR	Topic2 / LU2: frames and formats, structured prompting, state variants, reframing, grading and e
11:45–12:45	60	PP	LU2 practical: Labs02,03,04,05 core exercises, 15 minutes each; extensions completed after class
12:45–13:45	60	LUNCH	Lunch; excluded
13:45–15:00	75	CR	Topic3 / LU3: character sheets, local and global features, look bible, shot design (K5,K6,K7)
15:00–15:15	15	BREAK	Tea break; excluded
15:15–16:15	60	PP	LU3 practical: Labs06 and07, 30 minutes each (A3,A4)
16:15–17:30	75	CR	Topic4 / LU4: deep video models, shot prompting, continuity controls, cast detection and evaluat
17:30–18:30	60	PP	LU4 practical part1: Lab08 cast-presence detection and evaluation (A5)

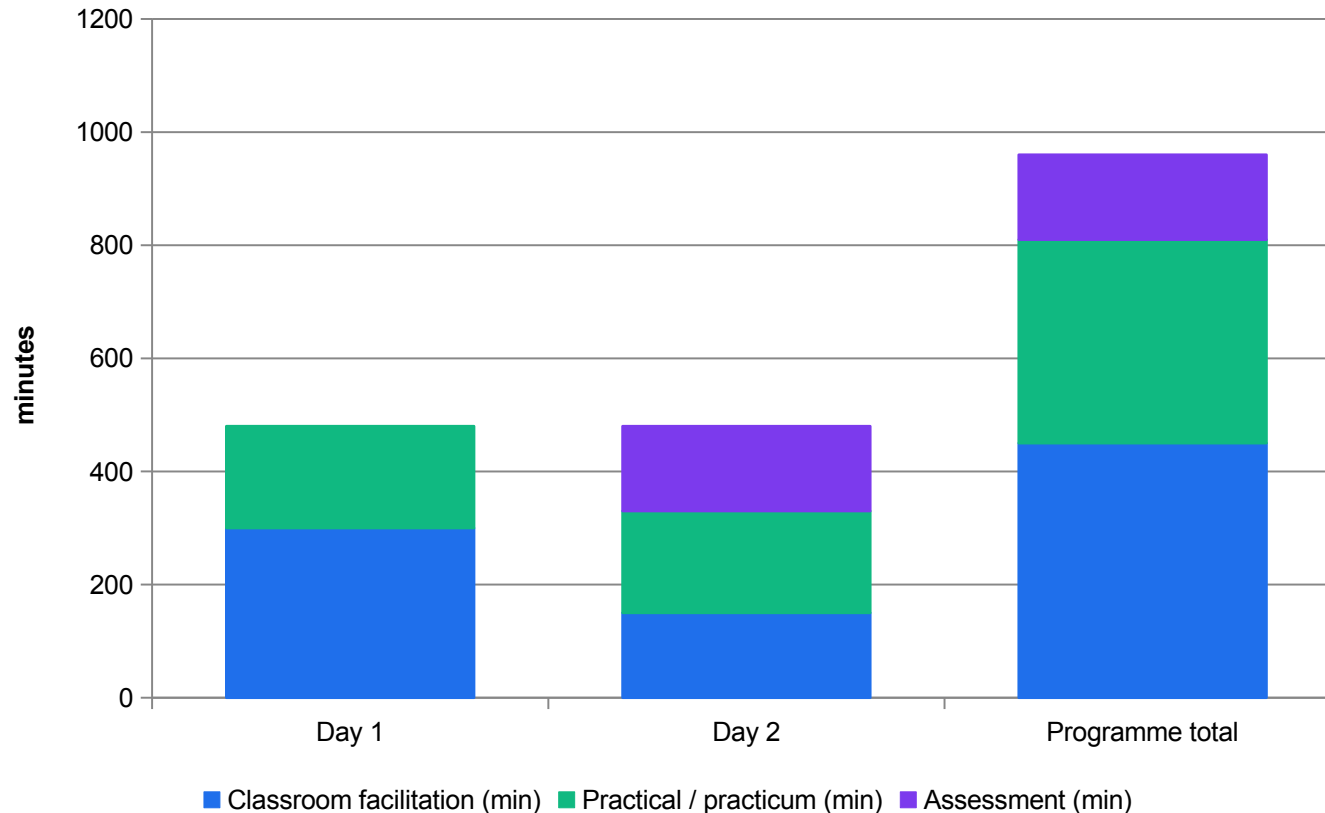
Programme minutes on Day 1: 480 (classroom facilitation + practical + assessment). Lunch, tea breaks and administration are additional, non-instructional time.

Lesson Plan — Day 2

Time	Min	Component	Session
08:45–09:00	15	ADMIN	Attendance and technical setup; outside programme minutes
09:00–10:00	60	PP	LU4 practical part2: Lab09 storyboard-to-shot prompts and Lab11 sound, captions and export, 30 m
10:00–10:15	15	BREAK	Tea break; excluded
10:15–11:30	75	CR	Topic4 / LU5: temporal coherence, cuts, motion, tracking, drift and event analytics (K10)
11:30–12:30	60	PP	LU5 practical: Lab10 shot, motion and continuity analytics (A6)
12:30–13:30	60	LUNCH	Lunch; excluded
13:30–14:45	75	CR	Topic5 / LU6: production architectures, protocols, constraints and cloud-versus-local trade-offs
14:45–15:00	15	BREAK	Tea break; excluded
15:00–16:00	60	PP	LU6 practical: Lab12 cloud, local and hybrid production architecture evaluation (A7,A8)
16:00–16:10	10	ADMIN	Assessment briefing and attendance; outside programme minutes
16:10–17:10	60	AS	Written Assessment: 13 open-ended questions, K1-K13, individual/open book
17:10–18:40	90	AS	Practical Performance: five tasks, A1-A8, individual/open book; submission and sign-off

Programme minutes on Day 2: 480 (classroom facilitation + practical + assessment). Lunch, tea breaks and administration are additional, non-instructional time.

Programme Hours — 16 Hours Reconciled



450 min classroom

7.5 hours across the five topics, 75 min per approved Learning Unit.

360 min practical

6 hours across 12 labs: 0 / 60 / 60 / 180 / 60 by topic.

150 min assessment

WA 60 min + PP 90 min = 2.5 h, per the registered Assessment Plan.

Breaks are additional

1-hour lunch, two 15-minute tea breaks and the administration block sit outside the 960 programme minutes.

MEASURED

Column totals taken from the approved course proposal: 7.5 h classroom facilitation + 6 h practical + 2.5 h assessment = 16 h (960 min). The Lesson Plan schedule sums to exactly these figures.

EVERY FOLDER IS INDEPENDENTLY USABLE

The 12 Hands-On Labs

La b	Title	Topic	A	K	In class
01	Film Needs Analysis and AI Production Pipeline Specification	01	A1	K1, K2	in LU1
02	Frames, Formats and Aspect Ratios for Film Delivery	02	A2	K3	15 min
03	Structured Prompting for Cinematic Stills	02	A2	K3, K4	15 min
04	State Variants, Clean-Up and Reframing with Masks	02	A2	K4	15 min
05	Cinematic Grading and Enhancement Measured with PSNR and...	02	A2	K4	15 min
06	Character Reference Sheets and Consistency Features	03	A4	K5, K6	30 min
07	Look Bible and Scene Continuity with Global Descriptors	03	A3	K7	30 min
08	Cast-Presence Detection and Detector Evaluation	04	A5	K8, K9	60 min
09	Storyboard to Shot Prompts, Start and End Frames, and the...	04	A5, A6	K8, K10	30 min
10	Shot, Motion and Continuity Analytics on the Assembled...	04	A6	K10	60 min
11	Voiceover, Music, Captions and the Final Export Ladder	04	A5, A6	K9, K10	30 min
12	Cloud, Local and Hybrid Production Architecture Evaluation	05	A7, A8	K11, K12, K13	60 min

Each lab folder carries its own numbered steps, data, local media, prompt pack, expected outputs, evidence checklist and troubleshooting. No lab needs a paid service; generation steps are optional extensions.

Entry Requirements and Who This Is For

1

Knowledge and skills

Computer literacy at ICAS Level 2; minimum 3 GCE 'O' Levels including English, or WPL Level 5.

2

Experience

Minimum 1 year of working experience; minimum 18 years of age.

3

Technical

Windows or Mac laptop, a modern browser, and Python 3 with opencv-python and NumPy. FFmpeg and AI tool accounts are optional.

4

Coding

Basic Python is helpful but not assumed — every lab script is supplied, run and explained step by step.

5

Typical roles

Content creator, digital marketer, social media manager, video editor, multimedia producer, brand and communications executive, trainer and instructional designer.

6

What you leave with

A planned, storyboarded and finished short film, a reusable prompt library, and quality checks for every shot you generate.

Briefing for Assessment

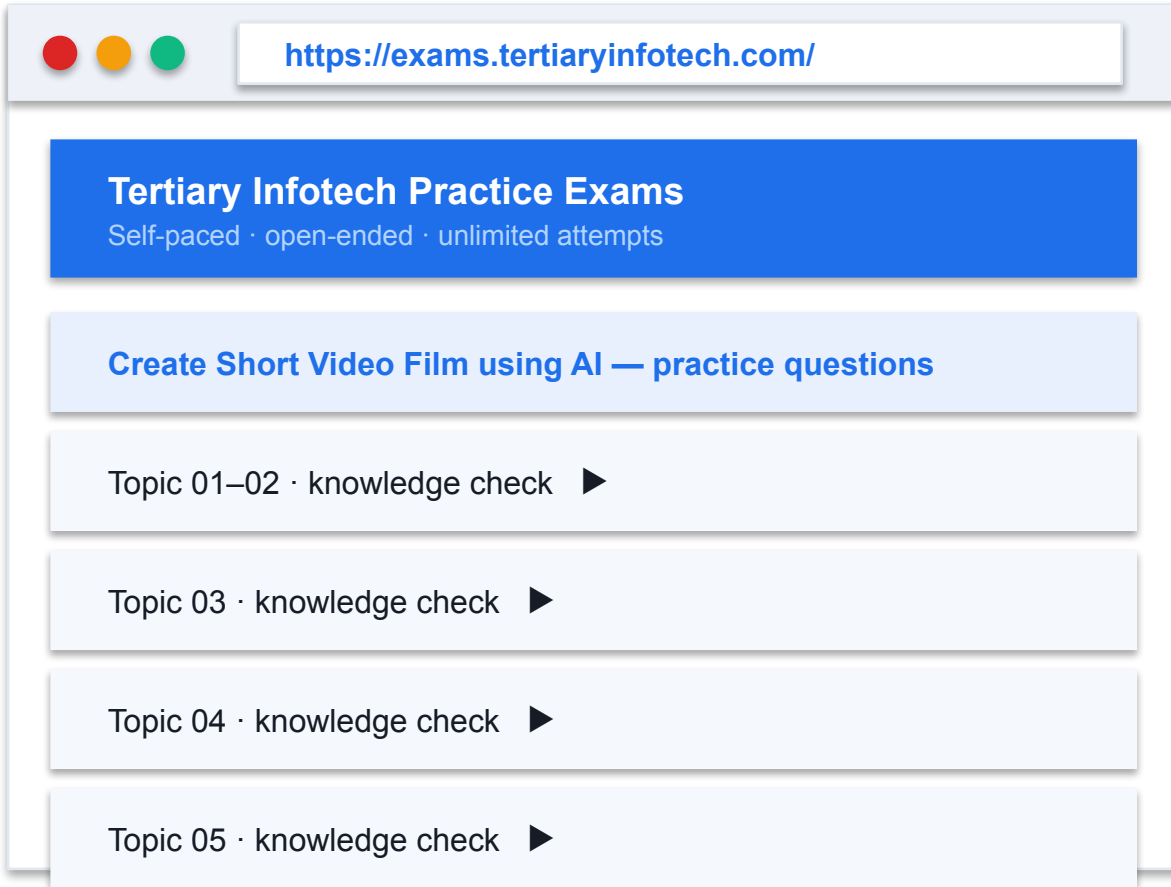
- Place phones and all other materials under the table or on the floor.
- No photographs and no recording of assessment scripts.
- No discussion during the assessment.
- Use a black or blue pen for any hard-copy response.
- No liquid paper and no correction tape.
- Assessment scripts are collected when time is up.
- Take the Assessment digital attendance before the paper starts.

Final Assessment

Instrument	Covers	Duration	Format
Written Assessment WA(Q&A)	K1 to K13 — 13 open-ended short-answer questions, one per knowledge statement	60 min	Individual, summative, open book
Practical Performance (PP)	A1 to A8 — 5 scenario-based tasks (Task 1: A1 · Task 2: A2, A3 · Task 3: A4 · Task 4: A5, A6 · Task 5: A7, A8)	90 min	Individual, summative, open book
Total	Full TSC coverage; assessor-to-candidate ratio 1:3 to 1:10	2.5 hours	Day 2, from 16:10

Open book means the slides, the Learner Guide and approved course materials only. A minimum 75% attendance record and a 'Competent' outcome are both required for funding. An appeal process is available.

Practice Exam

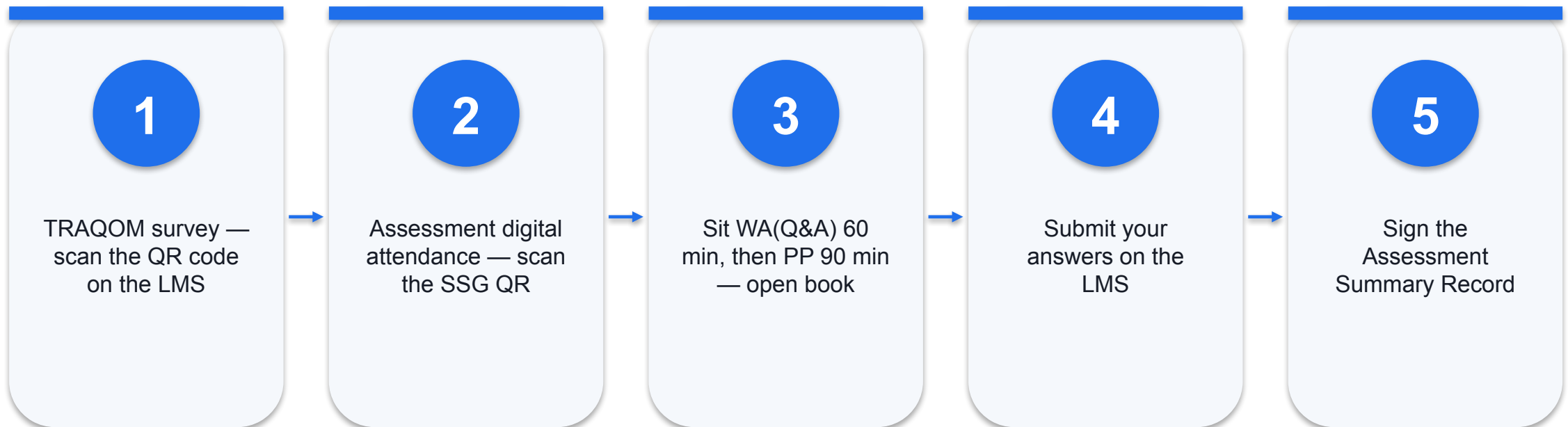


The screenshot shows a web browser window with the address bar displaying <https://exams.tertiaryinfotech.com/>. The main content area has a blue header with the text "Tertiary Infotech Practice Exams" and "Self-paced · open-ended · unlimited attempts". Below this is a light blue button labeled "Create Short Video Film using AI — practice questions". Underneath the button are four light gray buttons, each labeled "Topic 01–02 · knowledge check", "Topic 03 · knowledge check", "Topic 04 · knowledge check", and "Topic 05 · knowledge check", each with a right-pointing triangle icon.

- 1 Open the portal**
exams.tertiaryinfotech.com — sign in with your registered email.
- 2 Choose this course**
Select the practice set for this course.
- 3 Answer in your own words**
Open-ended questions, like the real Written Assessment.
- 4 Review**
Compare with the slides and Learner Guide; revisit weak topics.

Practice questions are for preparation only; they are not the assessment paper and are not graded.

Assessment Flow



Both instruments are open book and both are individual. The assessor records C or NYC against every knowledge and ability statement.

How Every Figure in This Deck Is Labelled

1

MEASURED

Timed or counted on stated hardware or data, with the sample size and the percentile given. Most numbers in this deck come from the labs you will run.

2

MODELLED

Derived from a documented relationship, with the relationship shown — for example diffusion latency proportional to S steps, or to $S \times F$ for F video frames.

3

PUBLISHED

Reported in a cited source, with the source and its page range named on the slide and in the source register.

4

ILLUSTRATIVE

Assumed values used to show the shape of a relationship. Never a quotation, never a forecast, and never to be lifted into a client document.

Provenance Rules for Every Asset You Touch

1

SIMULATED

The course illustrations of The Last Ferry: deterministic OpenCV drawings with a burned-in banner. Never described as a photograph or as model output.

2

AI-GENERATED

Kept unchanged with a record of the tool, model ID, date and exact prompt. Many generators also embed a watermark such as SynthID.

3

ANIMATIC

Storyboard keyframes timed to the voiceover and labelled on every frame. It fixes timing before credits are spent; it is never submitted as a generated film.

4

DERIVATIVES

A crop, reframe or grade of a labelled asset carries the same label. The label travels with the pixels.

TOPIC 01

AI Vision and Generative AI for Short Film Creation

The AI film pipeline as a vision system · how image and video generators work — diffusion, diffusion transformers, VAEs and GANs · text-to-image, image-to-video and text-to-video · reference-first production · business applications and the need test

Key Concepts — Topic 01

1

An AI film pipeline is a vision system with a synthesis stage

Sense (scripts, references, footage) -> represent (frames, latents, formats) -> analyse (continuity and quality measurements) -> decide (approve, regenerate, re-edit) -> synthesise (generate the next shot). Generation adds the last stage; it does not remove the first four.

2

Generators model $p(\text{frames} \mid \text{condition})$

Text-to-image, image-to-video and text-to-video differ only in the condition. The more the condition fixes — a character sheet, a location plate, a start frame — the less the model has to invent, and the less it drifts.

3

Today's video models are diffusion transformers

A VAE compresses frames to a latent; a transformer denoises spatiotemporal patches over S steps, guided by text and image conditions. GANs survive mainly as fast upscalers and face restorers. Each family fails in a recognisable way.

4

Reference-first beats prompt-first

Beginners send a story straight to a video model and reroll for hours. Professionals build character, location and prop references first, so the video model has nothing left to guess. Most of the quality is decided before the first clip is generated.

5

Business value comes from repeatable short-form storytelling

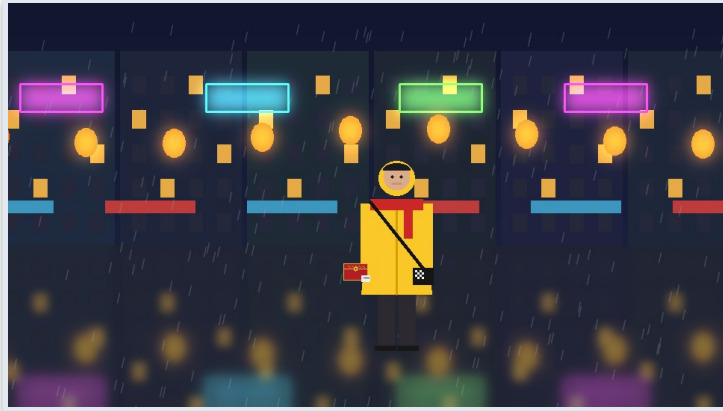
Brand films, product teasers, training micro-dramas, social series, pitch previsualisation and event recaps. Value comes from volume, speed to first cut and the ability to iterate a story without re-shooting.

6

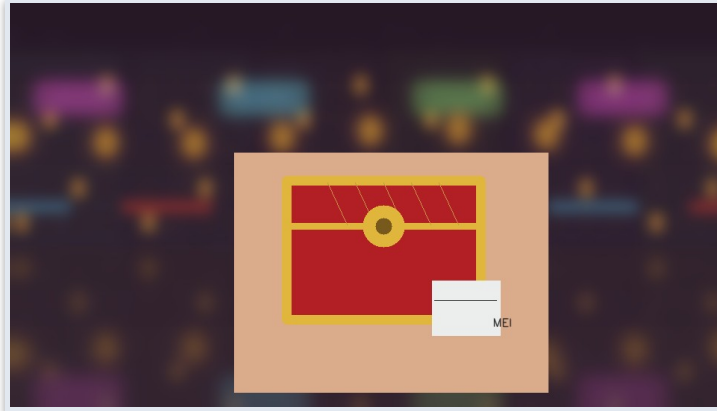
The need test decides the workflow, not the hype

Score each film idea on volume, repeatability, turnaround, review tolerance, likeness/rights risk and cost gap. Then choose: an all-in-one agent (fast, less control), a multi-tool pipeline (control, consistency) or a live shoot.

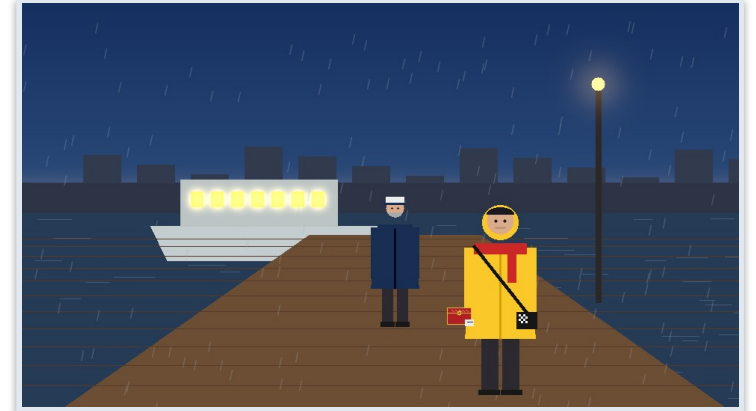
One Film, Twelve Labs: The Last Ferry



NIGHT MARKET — Mei races to the pier



RED BOX — the tag is still folded



HARBOUR PIER — Captain Tan waits

A young courier races through a rain-soaked night market to deliver a red box to the last ferry — and discovers it is addressed to her. Every lab builds a piece of this film. Illustrations are SIMULATED classroom stand-ins, not AI output.

An AI Film Pipeline Is a Vision System



SO WHAT Generation is only one stage. A film that nobody measured is a pile of clips; the analytical stages are what turn generations into a coherent film.

Why a Film Course Teaches Computer Vision

Understand the frame

- **Vision analytics: pixels → meaning**
 - Input: frames. Output: a label, a box, a mask, a number.
 - Detect the cast, track the subject, find the cuts.
 - Evaluated with IoU, precision, recall.
 - Fails visibly: a wrong count is checkable.

Create the frame

- **Generative vision: meaning → pixels**
 - Input: prompt, references, start frame. Output: frames.
 - Samples from $p(\text{frames} \mid \text{condition})$.
 - Evaluated with consistency, look and continuity measures.
 - Fails plausibly: a wrong frame still looks like a frame.

The last row is the reason: a generated shot that is wrong is not obviously wrong, so it has to be measured before it reaches the edit.

Three Generator Families You Will Meet

VAE

- Encoder compresses a frame to a small latent.
- Decoder turns latents back into pixels.
- Inside today's image and video models.
- Alone: smooth but soft output.
- Failure: lost fine detail.

GAN

- Generator vs discriminator game.
- Fast, sharp single-pass output.
- Now mostly upscalers and face restorers.
- Unstable to train.
- Failure: mode collapse, artefacts.

Diffusion / DiT

- Learns to remove noise over S steps.
- Transformer over spatiotemporal patches.
- Conditioned on text, images, frames.
- Today's image and video generators.
- Failure: morphing, drift, slow sampling.

Diffusion in Three Lines

Forward: add noise

```
x_t = sqrt(a_t) * x_0  
      + sqrt(1 - a_t) * e
```

Training takes a real clip x_0 and mixes in Gaussian noise e by a known schedule a_t .

Learn: predict the noise

```
loss = || e - model(x_t, t, c) ||^2
```

The network learns to predict the noise, given the step t and the condition c : prompt, reference images, start frame.

Sample: walk back

```
for t = S ... 1:  
  x_(t-1) = step(x_t, model(...))
```

Generation starts from pure noise and removes it over S steps. More steps and more frames cost more compute.

Video models run this in a VAE latent over spatiotemporal patches, so the same loop produces frames that agree with each other.

Generation Modes — What the Condition Fixes

Mode	Condition	What the model must invent	Use in the film
Text-to-image	Prompt	Everything	Explore looks cheaply
Image-to-image / edit	Prompt + image + mask	Only the masked region	State variants, clean-up, reframing
Reference-guided image	Prompt + element sheets	Pose, framing, light	Keyframes with the right cast
Image-to-video	Prompt + start frame (+ end frame)	The motion between	Most shots of the film
Text-to-video	Prompt only	Cast, place, look and motion	Rarely — least control
Video extension / reference	Prompt + previous clip	What happens next	Continuous action across a cut

The more the condition fixes, the less the model guesses — and the less it drifts.

THE REFERENCE-FIRST PRINCIPLE

Stop giving the model freedom.

Beginners take a story straight to a video model and reroll for hours. Build the character, location and prop references first: then the video model has nothing left to guess, and most of the work is done before the first clip.

Prompt-First versus Reference-First

Many rerolls

- **Prompt-first (what fails)**

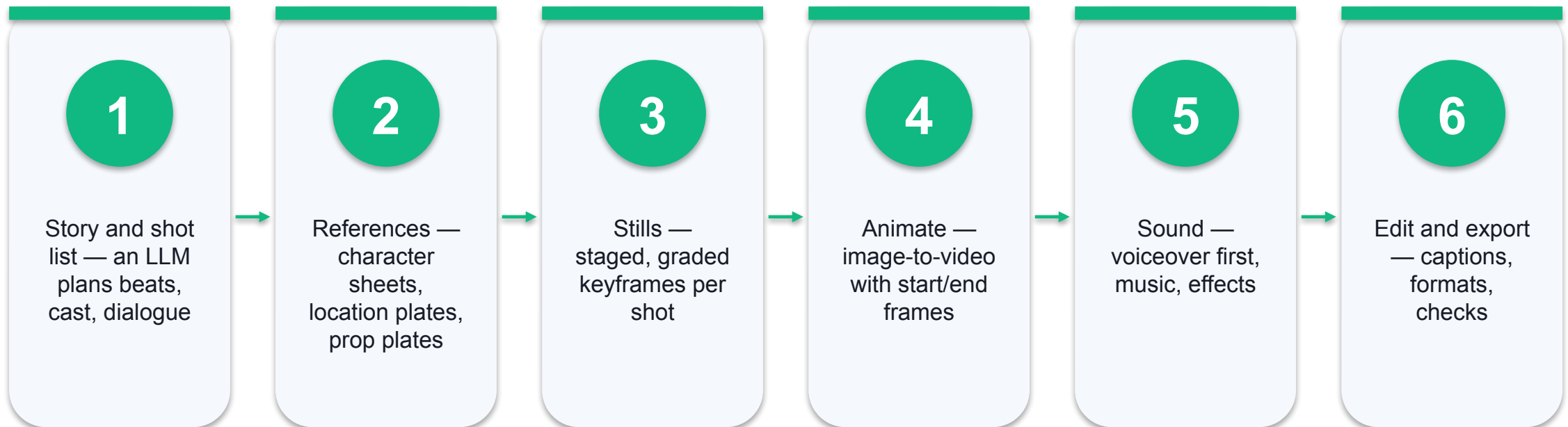
- Story → one long prompt → video model.
- The model invents the face, the coat, the pier.
- Every clip reinvents them slightly differently.
- Details in the prompt are forgotten.
- Dozens of rerolls; credits burned; no film.

Few takes

- **Reference-first (what works)**

- Story → shot list → references → stills → video.
- Character sheets, location plates, prop plates saved as named elements.
- Stills staged and graded before animation.
- Video model only adds motion.
- Few takes per shot; clips cut together.

The Production Workflow Used in Every Lab



Each arrow is a checkpoint. Nothing moves right until the measurements on the left have passed.

Business Applications of AI Short Film

Application	What AI changes	Watch-out
Brand short film	Hero film + cut-downs without a location shoot	Needs brand-hero quality and review
Product teaser / ad	Many variants of one concept for testing	Claims and brand safety
Social series	Same fictional hosts, weekly, fast	Consistency across episodes
Training micro-drama	Scenarios in many languages	Accuracy and regulator sign-off
Previsualisation	Animated storyboards before a live shoot	It is a plan, not the film
Event recap / explainer	Stylised re-enactment and motion graphics	Never fake a real event
Personalised video	One video per customer	Likeness, consent, PDPA

Value comes from volume, speed to first cut and cheap iteration — not from novelty.

The Six Need Criteria You Will Score

1

VOLUME

Finished minutes per quarter: 5 at 10+, 3 at 3–10, 1 below 3.

2

REPEATABILITY

Same cast, look and format every time scores 5; bespoke scores 1.

3

SPEED VALUE

Turnaround ≤ 5 days scores 5: fast iteration is where AI pays.

4

REVIEW TOLERANCE

5 where a person reviews every cut; 1 where output auto-publishes.

5

LIKENESS RISK

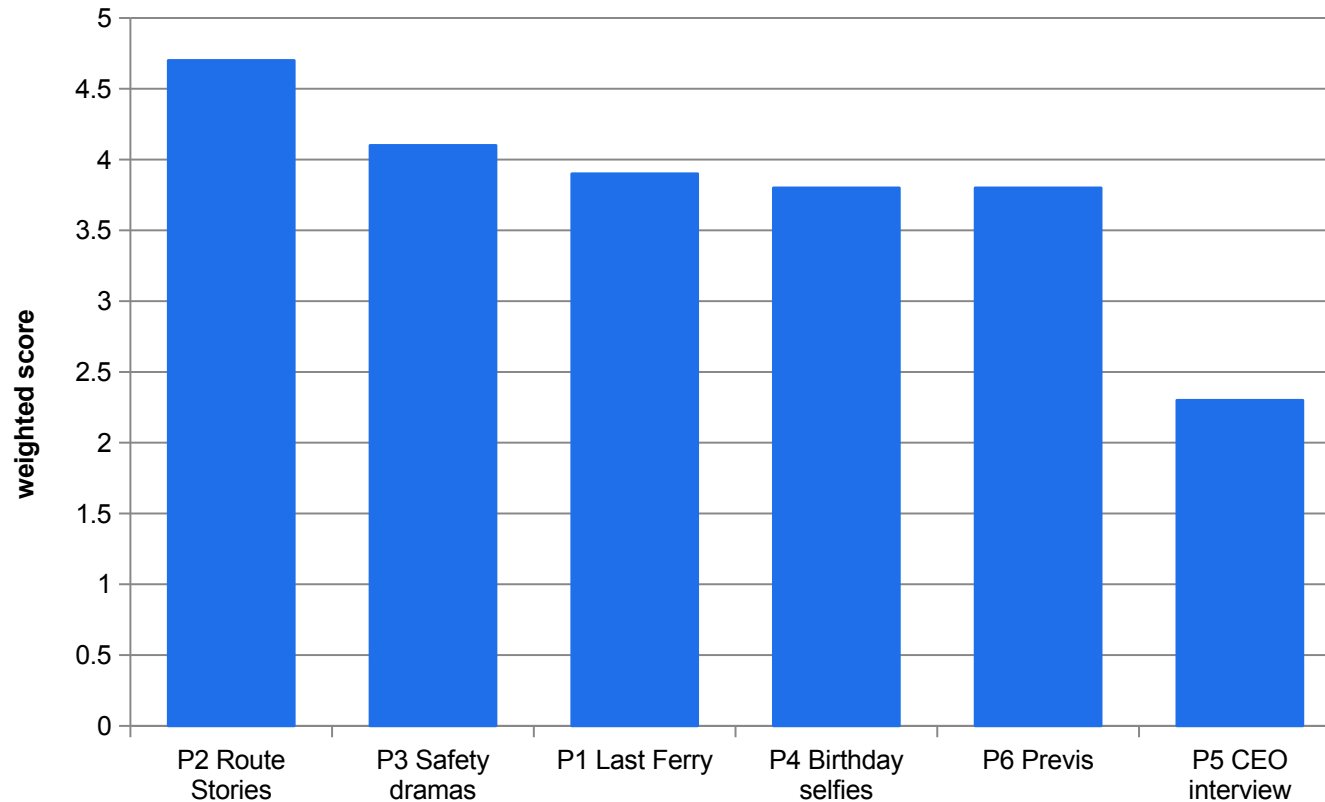
Inverted: 5 for fictional cast, 1 for unconsented real faces or voices.

6

COST GAP

(current – AI) $\div$ current: 5 at 0.6+, 3 at 0.3–0.6, 1 below.

Scoring the Six Harbourline Film Ideas



SELECT: P2, P1, P6

Above 3.6 and every gate passes.

HOLD: P3 (4.1)

Regulator approval pending — the gate is UNKNOWN.

REJECT: P4 (3.8)

Passes on score, fails the likeness gate.

REJECT: P5

A real executive's statement: AI is not appropriate.

MODELLED

Computed from the Lab 01 criteria and the fictional Harbourline data (labs/lab-01-film-needs-analysis). Threshold 3.6.

A1 · HOW TO READ A SCORING MATRIX

A likeness gate is a gate, not a weighting.

Apply consent, rights and authenticity FIRST. A project that scores 3.8 but generates passengers' faces without consent is rejected — no cost saving rescues it.

Three Workflow Patterns

All-in-one agent

- One photo + one structured prompt.
- Agent plans, generates, voices, scores.
- Minutes to a finished draft.
- Least shot-level control.
- Fits: social volume, quick drafts.

Multi-tool pipeline

- Planner → image → video → voice → music → editor.
- Named elements reused in every shot.
- Start/end frames, takes, grading.
- Most control; most steps.
- Fits: brand hero films.

Hybrid / local

- Cloud generation for hero shots.
- Local QA, grade, captions, mix, export.
- Optional open models on your GPU.
- Privacy and cost control.
- Fits: repeatable series.

The Tool Landscape by Production Stage

Stage	Job	Examples seen in current tutorials
Plan	Story, shot list, dialogue, prompts	Claude and other LLMs
References and stills	Character sheets, locations, props, keyframes	Nano Banana, GPT Image, Soul Cinema, Gemini image models
Storyboard	Arrange keyframes and references	Figma, node canvases (Flows)
Animate	Image-to-video, multi-shot, extension	Seedance, Veo, Kling, Cinema Studio
Voice	Voiceover, dialogue, audio tags	ElevenLabs, Qwen Audio
Music and effects	Score and sound design	Suno, ElevenLabs Music
Edit and finish	Timeline, captions, export	CapCut, Premiere Pro, DaVinci Resolve
All-in-one agents	Prompt + photo → drama	ZooClaw Vibe Drama, Higgsfield

Tool names come from the source tutorials (S6–S12) and change monthly. The course teaches the technique; check each tool's current terms before use.

Responsible AI Film-Making, From Day One

1

CONSENT AND LIKENESS

No real face or voice is generated, cloned or altered without written consent.

2

RIGHTS AND LICENCES

Check commercial-use terms for every model, voice and music track you ship.

3

PERSONAL DATA (PDPA)

Photos of customers or staff are personal data; they need a purpose and consent.

4

DISCLOSURE

Label AI-generated content on upload and in the credits.

5

PROVENANCE

Keep prompts, model IDs and dates; preserve C2PA credentials and watermarks such as SynthID.

6

NO DECEPTION

Never depict a real event, person or product claim that did not happen.

Lab 01 — Film Needs Analysis and AI Production Pipeline Specification

INPUTS

Concept key art for The Last Ferry (SIMULATED flat illustration, not AI output)

TECHNICAL QUESTION

Which film projects meet the weighted need criteria AND every gate — and which workflow suits each?

EVIDENCE TO PRODUCE

A scored shortlist with a written rationale for every rejected or held project, a workflow-pattern choice per selected project, and a five-stage pipeline specification for The Last Ferry listing the references it needs.

LAB 01 · IN-CLASS · Spreadsheet or text editor · Python 3 (optional baseline scorer) · no paid service required
Detailed procedures and prompts: [Learner Guide / labs/lab-01-film-needs-analysis/README.md](#)

TOPIC 02

AI Image Processing, Generation and Visual Enhancement

Frames, resolution, frame rate, aspect ratio and bitrate · structured prompting for cinematic stills · state variants, inpainting and outpainting for reframing · colour grading, denoising, sharpening and upscaling measured with PSNR and SSIM

Key Concepts — Topic 02

1

A frame is an H x W x C array; a film is frames plus a clock

OpenCV returns H x W x 3 uint8 in BGR order. A 1920x1080 RGB stream at 24 fps is $1920 \times 1080 \times 3 \times 24 = 149.3$ MB/s uncompressed; a delivery codec brings that to a few MB/s. Frame count / fps = duration.

2

Aspect ratio is a delivery decision made before generation

16:9 for YouTube and screens, 9:16 for Reels, Shorts and TikTok, 4:5 or 1:1 for feeds. A 9:16 centre crop of a 16:9 frame keeps only 31.6% of its width, so decide the master format first and keep action inside the shared safe area.

3

Structured prompts reduce variance

Six fields — subject, action, environment, camera, style, constraints — or the same fields as JSON. Lock the fields that must not change, vary one factor at a time, and explore on a fast, cheap model before committing credits.

4

An edit is masked, conditioned regeneration

State variants ('same pier, ferry pulling away — change nothing else'), inpainting to remove a stray object, outpainting to extend 16:9 to 9:16. White = edit, black = preserve in these labs. Always measure the pixels outside the mask: a good-looking edit can still move the background.

5

Grading is a point operation; filtering is a neighbourhood operation

White balance, curves and a teal-and-orange LUT map each pixel on its own. Blur, denoise and unsharp masking combine neighbouring pixels with a kernel. Resizing and reframing are geometric transforms (affine / perspective).

6

Enhancement must be measured, not admired

PSNR rewards per-pixel accuracy and forgives blur; SSIM rewards preserved structure. Upscale only the takes you keep, and check for halos and plastic skin before an enhancement reaches the timeline.

A Frame Is a Typed 3-D Array

```
>>> im = cv2.imread('reference/keyframe-shot2.png')
>>> im.shape, im.dtype
((540, 960, 3), dtype('uint8'))      # H, W, C
>>> im[515, 21]                       # [row, col] = [y, x]
array([255,  0,  0], dtype=uint8)     # pure BLUE: OpenCV is BGR
>>> cv2.cvtColor(np.uint8([[im[515,21]]]), cv2.COLOR_BGR2HSV)
array([[[120, 255, 255]]])           # hue 0-179 in 8-bit HSV
>>> cv2.imread('reference/nope.png')
None                                  # no exception: guard it
```

Shape is (H, W, C)

Height first. Index with [y, x]; draw with (x, y).

BGR, not RGB

Convert before plotting or sending to a model API.

uint8 wraps

$200 + 100 = 44$ in uint8. Convert to float for maths.

imread fails silently

It returns None; check before using the array.

VERIFY Run in Lab 02 — the four colour chips prove the channel order.

A Film Is Frames Plus a Clock

Duration

```
duration_s = frames / fps  
288 / 24 = 12.0 s
```

Count frames when it matters: containers can misreport the declared count.

Uncompressed rate

```
W x H x 3 x fps  
1920x1080x3x24 = 149.3 MB/s
```

Why raw frames never travel: a 48-second master would be 7.2 GB.

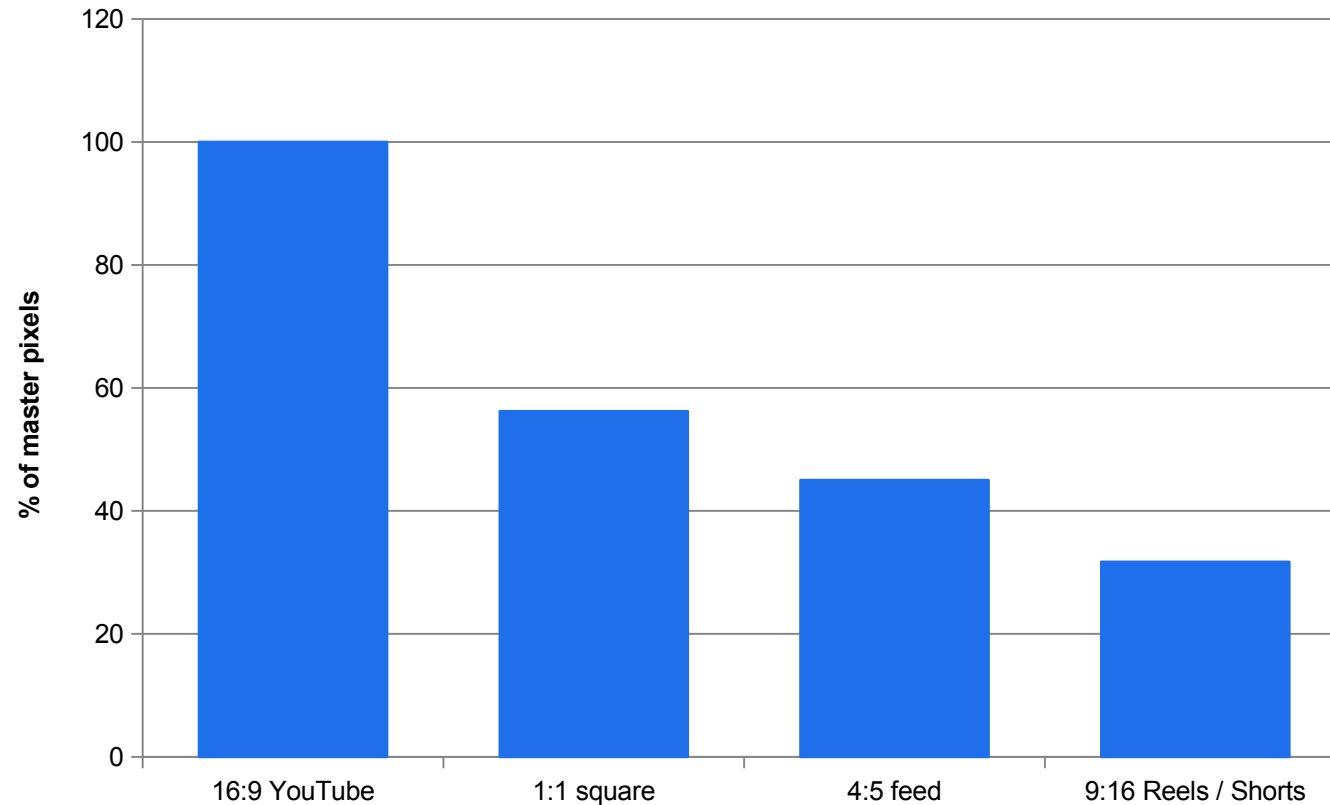
Bitrate

```
bitrate = bytes x 8 / duration  
4.87 MB -> 3.2 Mb/s
```

The classroom rough cut compresses about 92:1 against its raw rate.

Measured on the Lab 02 rough cut (960x540, 24 fps). A production master is 1920x1080 at the same aspect ratio.

What Each Delivery Format Keeps of a 16:9 Master



MEASURED

Centre crops of the Lab 02 master, computed by inspect_clip.py.

9:16 keeps under a third

Anything outside the centre third is lost.

Decide before generating

Native vertical, or keep action central.

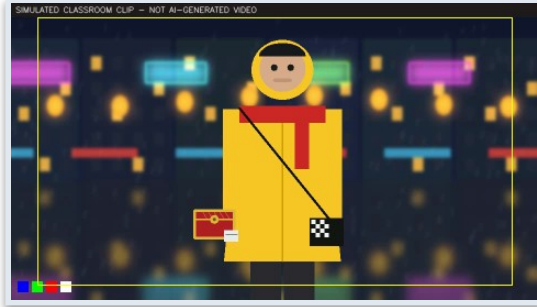
Mind the UI zones

Vertical apps cover ~14% top and ~20% bottom.

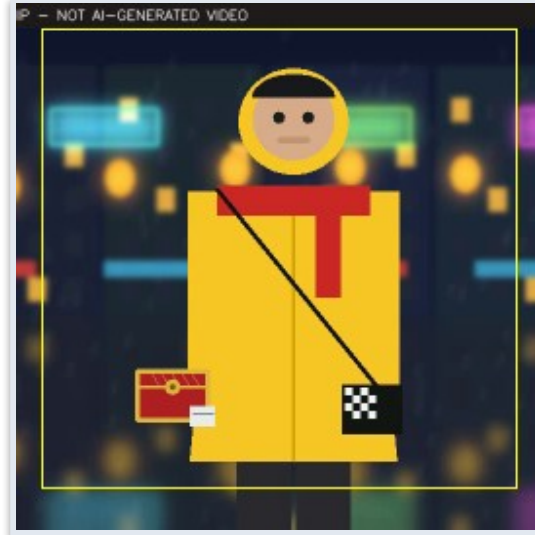
Upscaling is not free

A 9:16 crop needs about 3.6x to reach 1080x1920.

Reframing the Same Frame for Four Platforms



16:9 master



1:1



4:5



9:16

Blocking rule for every shot of The Last Ferry: faces, the red box and all captions stay inside the centre third of the 16:9 frame.

The Six-Field Shot Prompt

1

SUBJECT

Who, with the element reference: @mei in the yellow raincoat and red scarf.

2

ACTION

One clear action per shot: runs past neon signs, box under one arm.

3

ENVIRONMENT

The saved location: @night_market, rain, steam, wet ground.

4

CAMERA

Shot size, angle, lens, movement — described once.

5

STYLE

The look bible: blue hour, teal shadows, warm practicals, film grain.

6

CONSTRAINTS

What must not change: wardrobe colours, no extras, safe area.

Lint Before You Spend a Credit

```
$ python3 prompt_lab.py compose --fields data/shot2-fields-v1.json  
PROMPT (28 words)
```

```
A young woman courier. runs through the market holding a box.  
a busy night market at noon, neon signs. close-up wide shot.  
cinematic, cinematic, cinematic film look.
```

LINT

- MISSING field 'constraints' - the model will invent it
- CONFLICT 'night' vs 'noon' - pick one
- CONFLICT 'close-up' vs 'wide shot' - pick one
- REPEATED ['cinematic'] - repetition dilutes
- NO references - attach the character sheet and plate

Missing field = invented

Anything you leave out, the model decides for you.

Contradictions

Night and noon cannot both win; one silently loses.

Repetition

Saying it three times does not emphasise it.

References first

@elements go at the front of every prompt.

The Same Shot as a Versioned JSON Prompt

```
{
  "prompt_version": "shot2-v2",
  "references": ["mei_sheet_v1", "night_market_plate_v1", "red_box_v1"],
  "subject": "MEI: yellow hooded raincoat, red scarf, black bag ...",
  "action": "runs past neon signs, the red box under one arm",
  "environment": "night market, light rain, steam, wet ground",
  "camera": "medium shot, eye level, 35mm, tracking left to right",
  "style": "blue hour, teal shadows, warm lanterns, 35mm grain",
  "constraints": "coat and scarf exactly as the sheet; no extras",
  "negative": ["extra people in foreground", "text", "daylight"],
  "aspect_ratio": "16:9", "resolution": "1K draft", "seed": 4127
}
```

Versioned

shot2-v2 can be diffed, reviewed and rolled back.

Reusable

Swap one field to make the next shot.

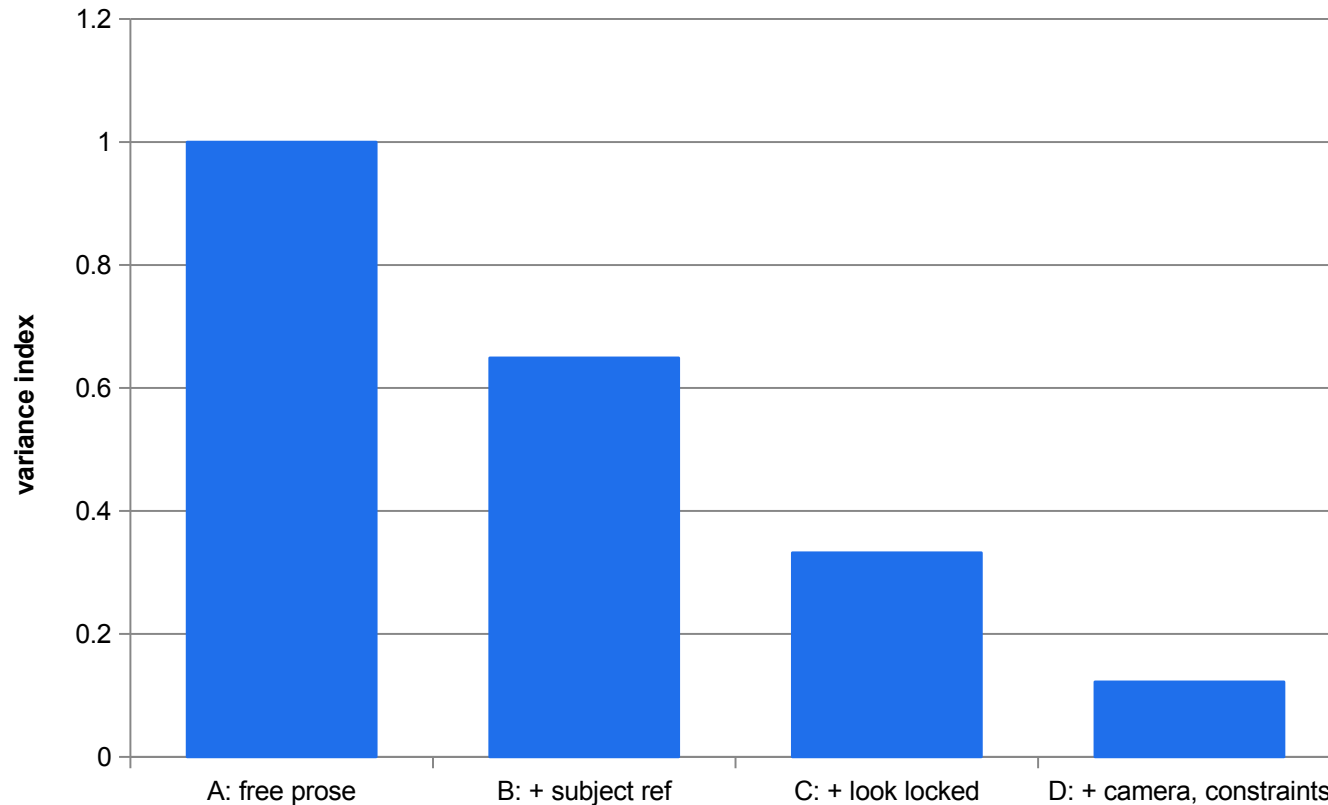
Draft first

1K draft on a fast model; upscale only keepers.

Rewriting off

Turn automatic prompt rewriting off for long prompts.

Locking Fields Reduces Take-to-Take Variation



Subject reference

Removes most wardrobe (coat hue) drift.

Look locked

Removes most background warmth drift.

Camera + constraints

Removes blocking and framing drift.

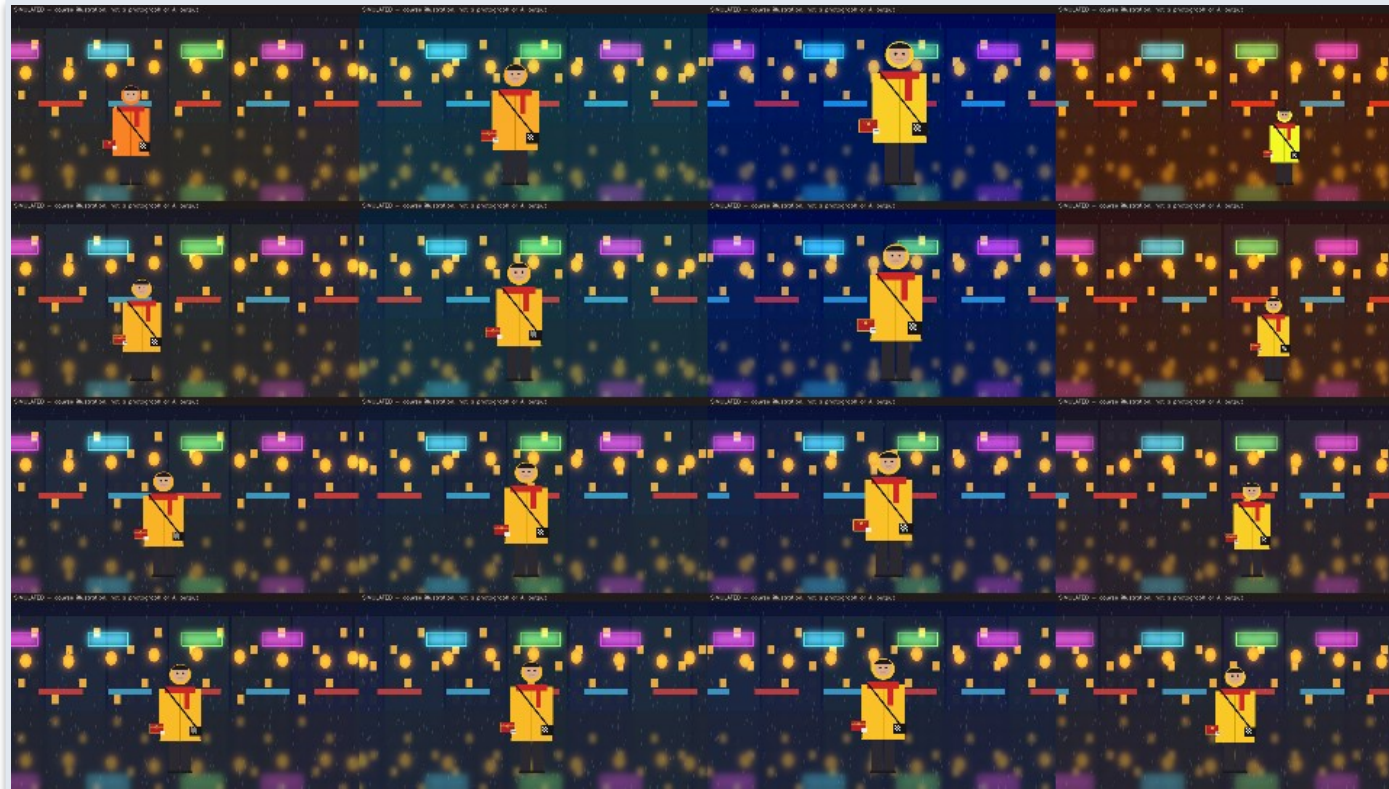
Leave some freedom

Crowd and steam may vary; the cast may not.

MEASURED

Measured by prompt_lab.py on 16 SIMULATED takes with a designed spread; real model output is noisier — rerun on your own takes.

Sixteen Takes, Four Prompt Designs



Rows A→D: each row locks more fields; columns are four takes

Row A

Coat colour, position and size all wander.

Row B

Coat fixed by the sheet; framing still drifts.

Row C

Look fixed; blocking still moves.

Row D

Four near-identical takes: pick on performance.

Explore Cheap, Commit Late

1

DRAFT MODELS

Explore looks on a fast, cheap image model at 1K before any high-quality generation.

2

VARIATIONS

Ask for four variations of one prompt, not four prompts.

3

STILLS BEFORE VIDEO

A still costs a fraction of a clip. Approve the frame, then animate it.

4

UPSCALE KEEPERS ONLY

Generate at low resolution; upscale only approved takes.

Editing Is Masked, Conditioned Regeneration

Inpaint

```
mask = WHITE on the object  
edit inside, keep outside
```

Remove the stray cone; the planks are continued from their surroundings.

State variant

```
same plate + 'change nothing else'  
+ allowed zone
```

Same pier, ferry pulling away. Verify every pixel outside the zone.

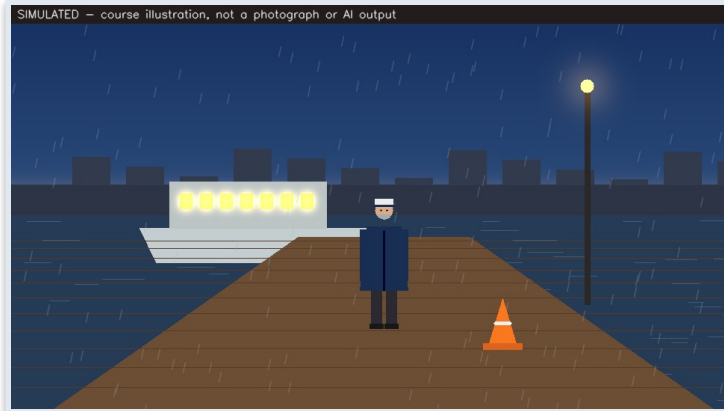
Outpaint

```
canvas 16:9 -> 9:16  
mask = WHITE on new area
```

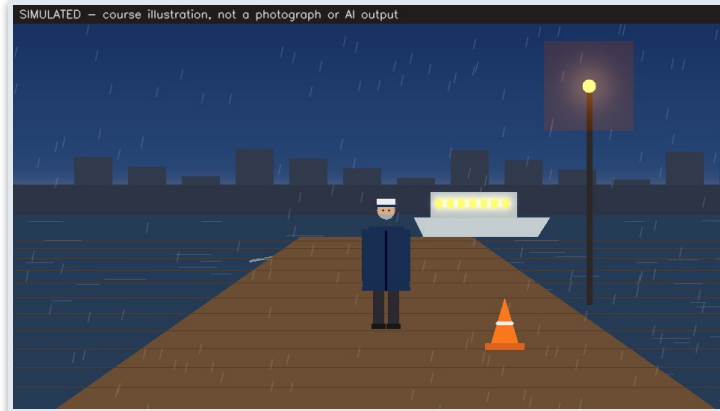
Keeps 100% of the source but must generate ~70% of the pixels.

Mask convention in this course: WHITE = edit, BLACK = preserve. Some tools use the opposite — always check.

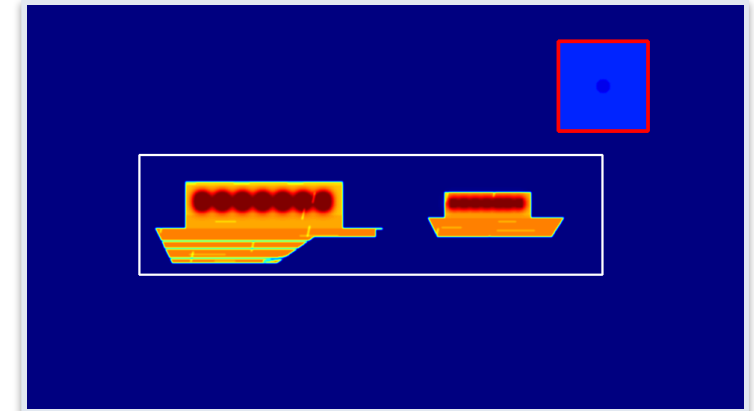
'Change Nothing Else' Is a Request, Not a Guarantee



State A — the plate



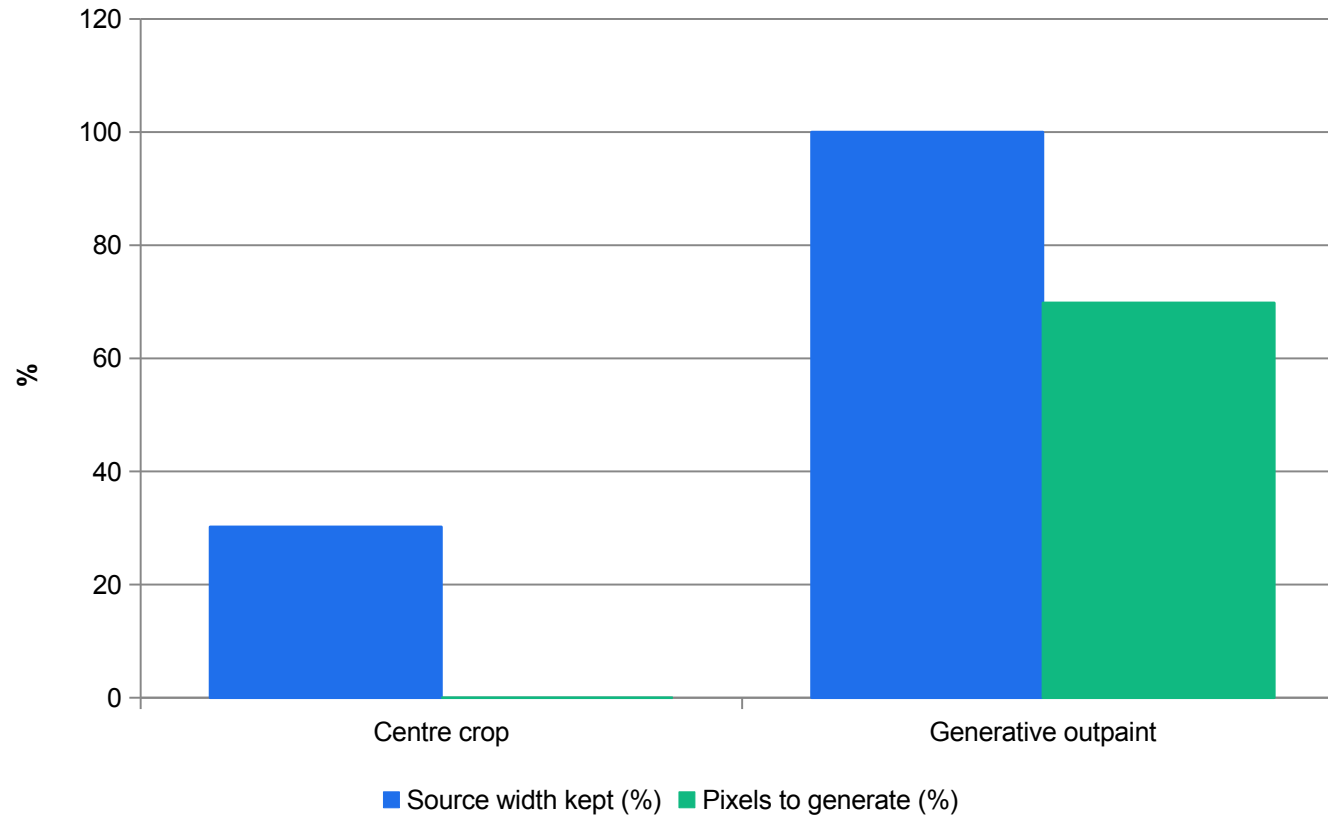
State B — as the model returned it



Difference: white box = allowed, red = unrequested

The ferry change sits inside the allowed zone. The lamp glow also changed (region [711, 48, 120, 120]): verdict REGENERATE, or fix the lamp with a second masked edit.

Crop or Outpaint for the 9:16 Cut-Down?



Crop

Nothing invented; two-thirds of the frame lost.

Outpaint

Whole frame kept; most of the picture is new.

Verify the new area

Outpainted sky and pier must match the look.

Or generate natively

A 9:16 generation avoids both costs.

MEASURED

Computed by edit_lab.py reframe on the 960x516 pier plate (banner removed).

Point Operations versus Neighbourhood Operations

Grade

- **Point operation: pixel → pixel**
 - Each output pixel depends only on the same input pixel.
 - White balance, exposure, curves, levels.
 - A LUT: one lookup table per channel.
 - The teal-and-orange look of the film.
 - Cannot sharpen, denoise or remove blur.

Filter

- **Neighbourhood operation: kernel**
 - Each output pixel combines its neighbours.
 - Blur, denoise (bilateral, non-local means).
 - Sharpen: unsharp mask, Laplacian kernel.
 - Resize: nearest, cubic, Lanczos.
 - Changes structure — so it can create halos.

Measuring an Enhancement

Unsharp mask

```
out = (1 + a) * img  
      - a * blur(img, s)
```

Adds back the detail a blur removes. Too large an amount a creates halos.

PSNR

```
10 * log10(255^2 / MSE)
```

Per-pixel error in dB. Rewards smoothing, forgives lost texture.

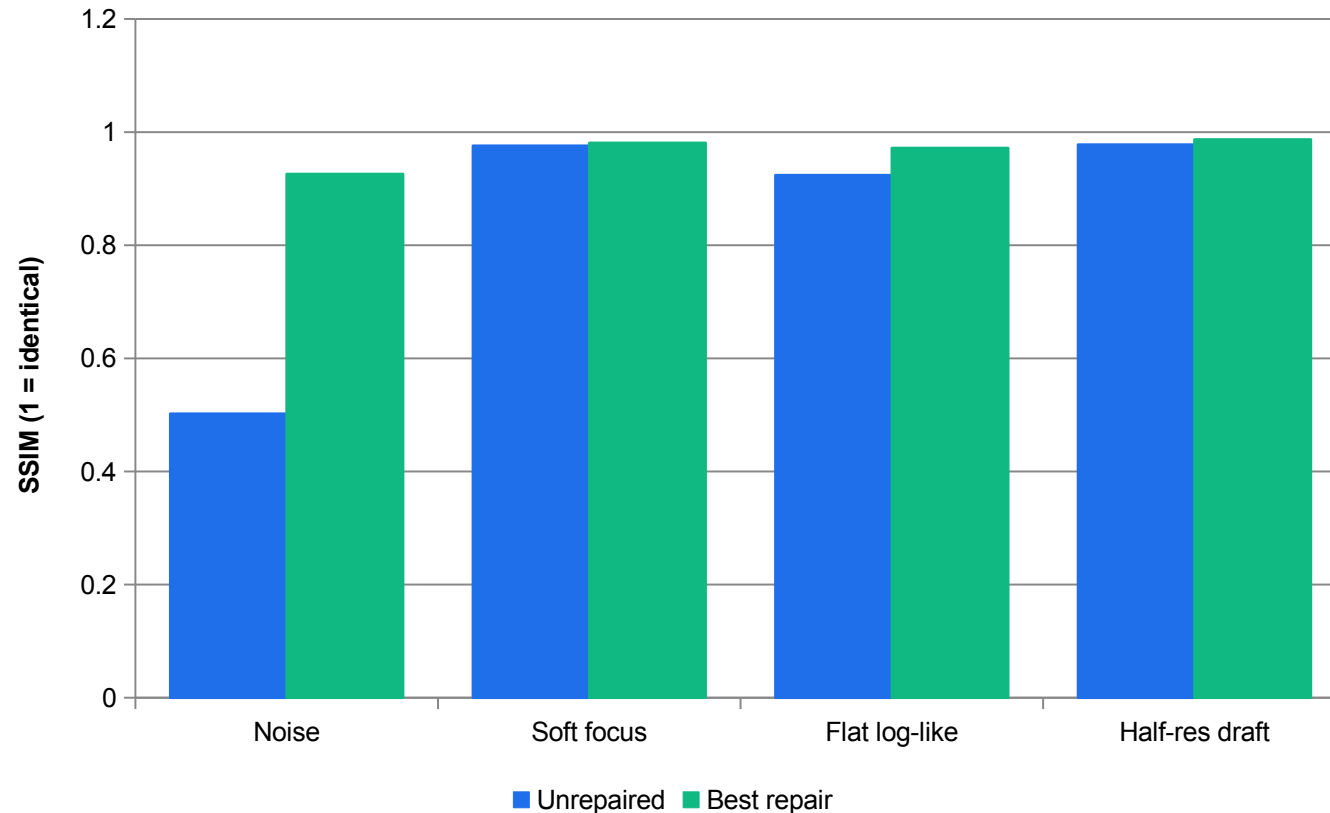
SSIM

```
[l * c * s] over 11x11 windows
```

Luminance, contrast and structure. Punishes lost texture PSNR forgives.

cv2.quality (SSIM) is contrib-only; the course ships a NumPy SSIM in ssim.py.

Best Repair per Defect — SSIM Against the Master



Noise

The biggest gain: 0.50 → 0.93.

PSNR disagrees

PSNR prefers bilateral (34.6 dB); SSIM prefers Gaussian.

Over-sharpen

Unsharp 2.5 wins PSNR, loses SSIM: halos.

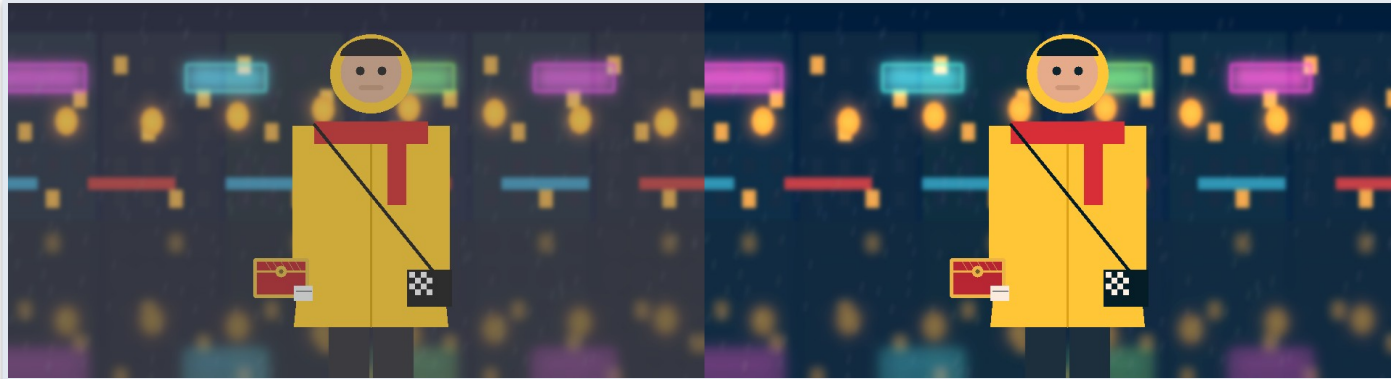
Over-clip

Levels 60-190 collapses SSIM to 0.40.

MEASURED

grade_lab.py on the SIMULATED graded master of shot S2. Best repairs: Gaussian blur, unsharp 0.8, levels 36-214, cubic x2.

Grading the Look as a Point Operation



Left: flat, log-like draft · Right: levels + teal-and-orange LUT

Shadows cooler

Red minus blue in the shadows falls.

Highlights warmer

Lanterns and skin gain warmth.

Grade stills first

Animate the graded still; the clip inherits the look.

Measure it

The warmth numbers go in the look bible (Lab 07).

Lab 02 — Frames, Formats and Aspect Ratios for Film Delivery

INPUTS

Keyframe of shot S2 with four exact colour chips bottom-left (SIMULATED)

TECHNICAL QUESTION

What does each delivery format keep of the 16:9 master, and what does the payload cost?

EVIDENCE TO PRODUCE

A clip report (shape, dtype, BGR/RGB/HSV of the calibration chips, fps, frame count, duration, bitrate, compression ratio), a contact sheet, and a reframe per delivery format with the percentage of the master each one keeps.

LAB 02 · 15 MIN · Python 3 · opencv-python · NumPy · no network

Detailed procedures and prompts: [Learner Guide / labs/lab-02-frames-formats-aspect-ratios/README.md](#)

Lab 03 — Structured Prompting for Cinematic Stills

INPUTS

16 takes of shot S2: four prompt designs × four takes (SIMULATED, designed spread)

TECHNICAL QUESTION

How much does each locked prompt field reduce the measured variation between takes?

EVIDENCE TO PRODUCE

A linted v1 prompt with its problems listed, your corrected v2 prompt in prose and JSON, and a variance table showing how much each prompt design reduced wardrobe, blocking, framing and look drift.

LAB 03 · 15 MIN · Python 3 · opencv-python · NumPy · optional: any image generator for your own takes
Detailed procedures and prompts: [Learner Guide / labs/lab-03-structured-prompting-stills/README.md](#)

Lab 04 — State Variants, Clean-Up and Reframing with Masks

INPUTS

HARBOUR PIER state A with a stray orange cone (SIMULATED)

TECHNICAL QUESTION

Which pixels changed inside and outside each edit mask, and did the edit obey 'change nothing else'?

EVIDENCE TO PRODUCE

An inpainted plate with proof that nothing outside the mask moved, a verification heat-map and verdict for the state edit, and a 9:16 crop and outpaint mask with the percentage of pixels each approach keeps or must generate.

LAB 04 · 15 MIN · Python 3 · opencv-python · NumPy · optional: an image editor or generative edit tool
Detailed procedures and prompts: [Learner Guide / labs/lab-04-state-variants-reframing/README.md](#)

Lab 05 — Cinematic Grading and Enhancement Measured with PSNR and SSIM

INPUTS

Approved graded master of shot S2
(SIMULATED)

TECHNICAL QUESTION

Which enhancement best repairs
each defect by PSNR and SSIM,
and do the two metrics agree?

EVIDENCE TO PRODUCE

An enhancement score table (4
defects × 3 repairs + unrepaired), a
before/after sheet of the best repair
per defect, and a flat-versus-
graded comparison with the
measured shift in shadow and
highlight warmth.

LAB 05 · 15 MIN · Python 3 · opencv-python · NumPy · no network

Detailed procedures and prompts: [Learner Guide / labs/lab-05-grading-enhancement-metrics/README.md](#)

TOPIC 03

AI Visual Features, Character Consistency and Scene Design

Features as the language of consistency · character reference sheets and AI casting · local features — colour, edge, texture and keypoints · global features — histograms, statistics and moments for the look of a shot · shot size, camera angle, movement and lighting

Key Concepts — Topic 03

1

Consistency is a feature-matching problem

A character is 'the same' when repeatable features match across shots: wardrobe colours, silhouette, facial landmarks, a distinctive prop. The video model conditions on those features, and your QA measures them.

2

A character reference sheet is a feature specification

Front, side and back views plus close-ups and expressions, one outfit, fixed colours. Casting attributes — genre, era, archetype, identity, build, details, outfit — pin the look before the first shot is generated.

3

Local features answer 'what is here?'

Dominant colour in HSV, Canny edges of the silhouette, texture energy and ORB keypoints on the face or prop. They are compared crop to crop, so they catch a scarf that changed colour or a prop that lost its clasp.

4

Global features answer 'does this whole shot belong?'

HSV histograms compared with Bhattacharyya distance, mean luminance, contrast, warm/cool balance and Hu moments of the composition mask. Together they form a numeric 'look bible' that every shot must stay close to.

5

Scene design is composition, camera and light

Shot size (wide, medium, close-up), angle (eye-level, low, high, over-the-shoulder), movement (push-in, pan, tracking, handheld) and lighting (key, fill, rim, practicals, blue hour). The rule of thirds and the 180-degree rule keep a cut readable.

6

Locations and props need angles and states

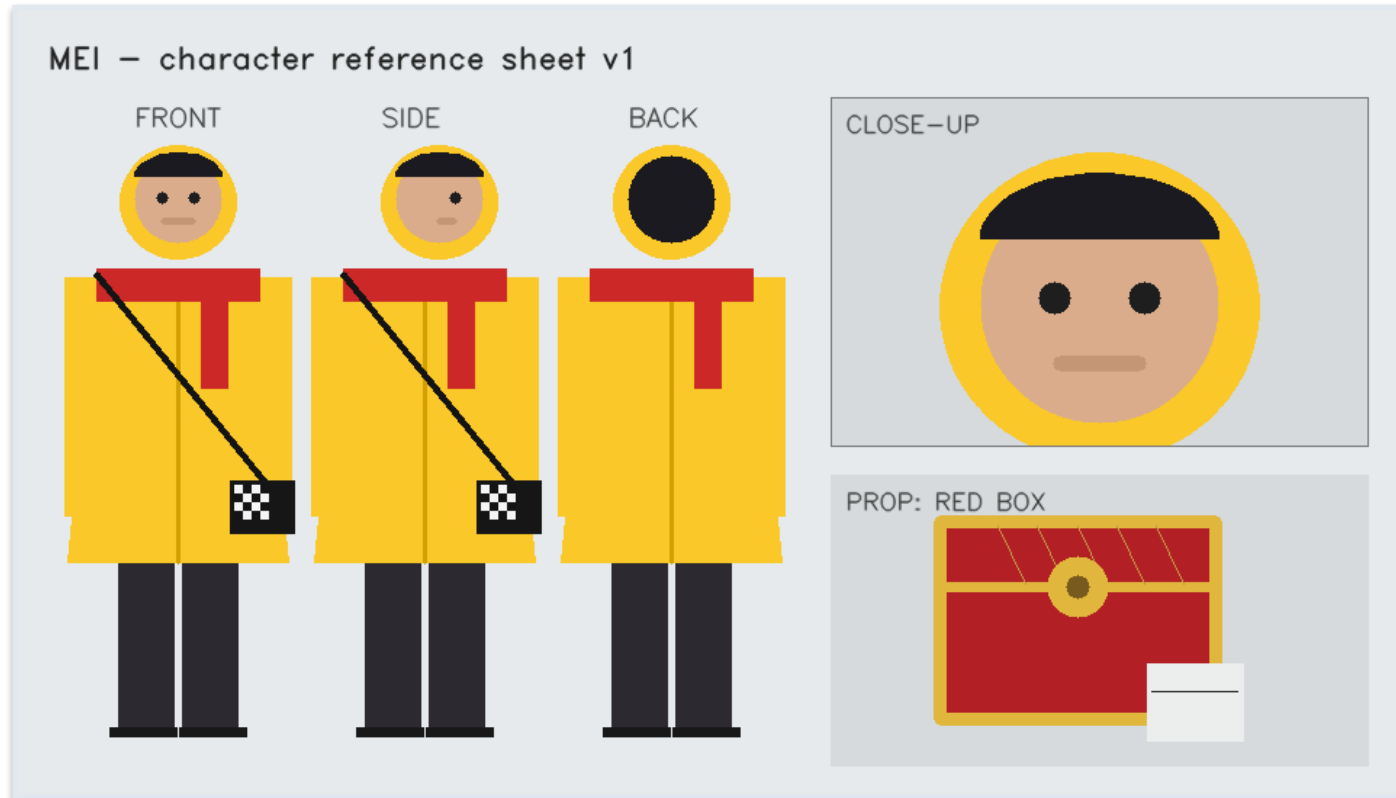
Generate a multi-angle location grid and each story state of a location (ferry docked, ferry leaving) as its own reference, plus a clean hero plate of every prop. The model reuses what you give it and invents what you don't.

K5 · FEATURE EXTRACTION AND REPRESENTATION

Consistency is a feature-matching problem.

A character is 'the same' across shots when repeatable features match: wardrobe colours, silhouette, a face, a distinctive prop. The generator conditions on those features — and your quality check measures them.

A Reference Sheet Is a Feature Specification



MEI — front, side, back, close-up and the RED BOX prop

Three views

Front, side and back fix the silhouette from every angle.

Close-up

Face, scarf and the chequered bag patch at full detail.

One outfit

Fixed colours: yellow raincoat, red scarf, black bag.

Save as an element

Name it (@mei) and reuse it — never re-describe her.

Casting a Character Without a Photo

1

GENRE

Sets the visual register: drama, action, thriller, romance.

2

BUDGET AND ERA

Production scale and period shape wardrobe and grooming.

3

ARCHETYPE

The role in the story: hero, sage, innocent, rebel.

4

IDENTITY

Gender, age and ethnicity chosen deliberately, not left to chance.

5

BUILD AND DETAILS

Body type, height, facial hair, scars, tattoos.

6

OUTFIT

One costume, specified exactly — it is the strongest feature.

Generate, Approve, Save, Reuse



The model then loads the same approved reference every time instead of reading a fresh description and guessing.

Local Features and the Failure Each One Catches

Feature	How it is computed	Catches	Blind to
Coat hue	Median OpenCV hue inside the coat mask	Wardrobe colour drift	Shape changes
Scarf red cover	Share of red pixels in the upper body	Scarf changed or missing	Which red
Edge density	Canny edge pixels / crop pixels	Silhouette and detail changes	Colour
Texture energy	Variance of the Laplacian	Plastic, over-smoothed faces	Identity
ORB matches	Keypoints + Hamming + ratio test vs the sheet	A different patch, face or prop	Colour: it runs on grey
Clasp window	Brass pixels inside the clasp box only	Lost prop detail	Anything outside the window

No single feature sees everything. The check is a designed combination — that design is ability A4.

Matching Keypoints to the Reference Sheet

```
orb = cv2.ORB_create(nfeatures=800)
ka, da = orb.detectAndCompute(sheet_gray, None)
kb, db = orb.detectAndCompute(shot_gray, None)
pairs = cv2.BFMatcher(cv2.NORM_HAMMING).knnMatch(da, db, k=2)
good = [m for m, n in pairs if m.distance < 0.75 * n.distance]

hsv = cv2.cvtColor(crop, cv2.COLOR_BGR2HSV)
coat = cv2.inRange(hsv, (5,150,170), (40,255,255)) > 0
hue = np.median(hsv[..., 0][coat])          # sheet: 23
```

ORB descriptors are binary

32 bytes per keypoint; compare with Hamming.

Ratio test

Keep a match only if clearly better than the runner-up.

Resize crops first

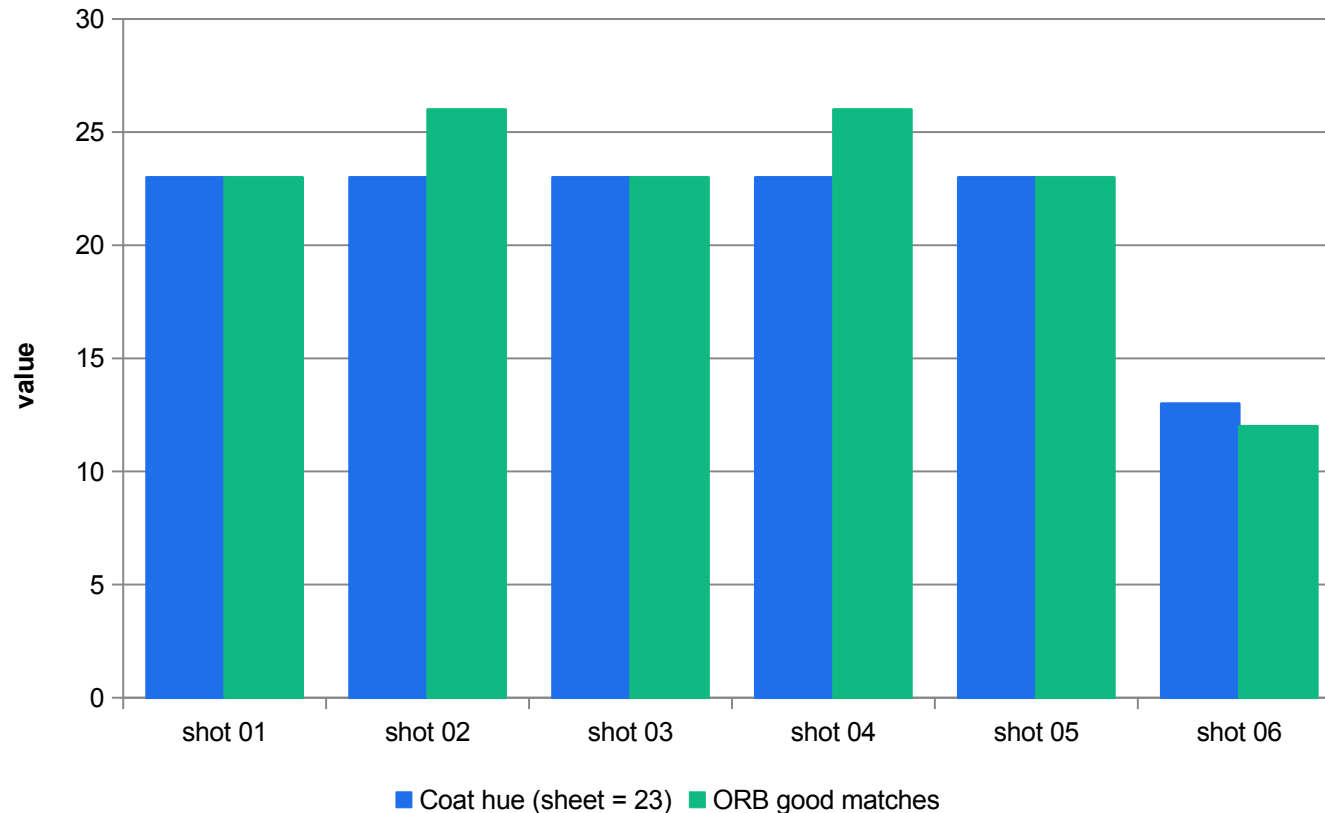
Bring both to the same height so scale is comparable.

Hue in OpenCV

0–179 in 8-bit: yellow sits near 22–23.

VERIFY character_features.py in Lab 06 runs this on six shots.

Six Shots of MEI Against Her Reference Sheet



shot 04: DRIFT

Scarf red cover 0.00 — the scarf turned blue.

shot 05: DRIFT

Clasp brass 0.14 — the clasp is missing.

shot 06: DRIFT

Coat hue 13, not 23 — the coat drifted to orange.

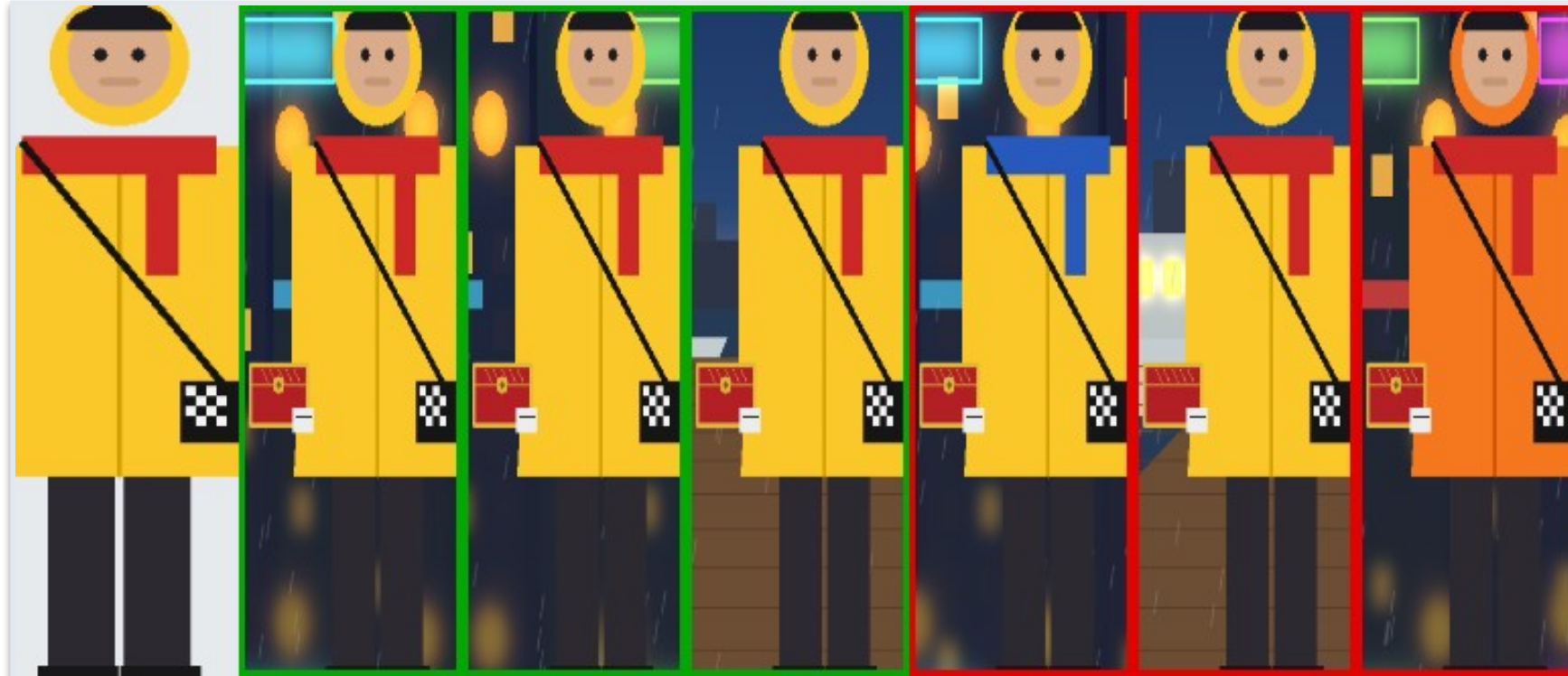
Keypoints miss colour

shot 06 still gets 12 ORB matches.

MEASURED

character_features.py on SIMULATED shots with injected drift.

The Identity Strip



Sheet crop first, then shots 01–06 · green = consistent, red = drift

Regenerate shots 04, 05 and 06 with the sheet re-attached and the drifted feature named in the constraints.

Local versus Global Features

Lab 06

- **Local: what is HERE?**
 - Computed on a crop or a point.
 - Edges, corners, colour patches, ORB keypoints, optical flow.
 - Answers: is this still MEI? is the clasp there?
 - Used for character and prop consistency.

Lab 07

- **Global: what is this WHOLE shot like?**
 - Computed on the whole frame.
 - Histograms, mean brightness, contrast, warmth, tint, moments.
 - Answers: does this shot belong in this film?
 - Used for the look bible and scene continuity.

A Numeric Fingerprint for the Look

Histogram distance

```
d_B(H1, H2) =  
sqrt(1 - sum sqrt(H1*H2))
```

Bhattacharyya distance of normalised hue/saturation histograms: 0 = same palette.

Statistics

```
mean V, std V, mean S  
warmth = mean(R - B)  
tint = mean(G - (R+B)/2)
```

Exposure, contrast, saturation, white balance and green/magenta cast.

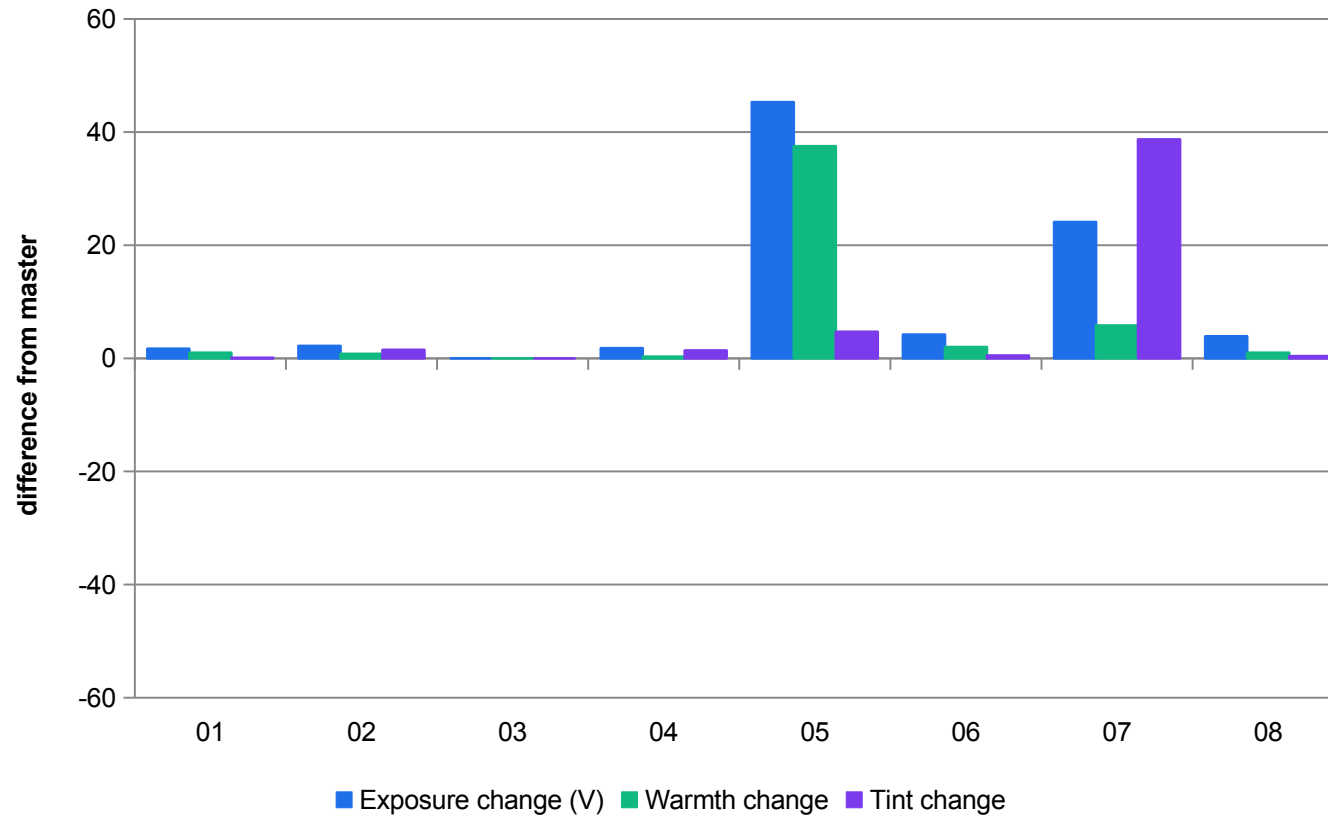
Geometry

```
Hu moments of the  
bright-practicals mask
```

Where the light sits in the frame — invariant to position and scale.

Each shot is compared with the approved master of ITS OWN location.

Auditing Eight Shots Against the Look Bible



shot 05 OFF LOOK

Exposure +45, warmth +38: a 'noon' generation.

shot 07 OFF LOOK

Tint -39, saturation +54: a magenta cast.

Six shots ON LOOK

Every delta well inside the limits.

Regrade or regenerate?

A LUT can fix a cast; not daylight.

MEASURED

look_audit.py: each shot minus its location master. Limits: exposure 15, warmth 20, tint 12.

The Look Strip



Shots 01–08 · green = on look, red = off look

Compare with one master for the whole film: every market shot would be flagged — a single master punishes legitimate differences between locations.

Shot Size and Camera Angle Vocabulary

Term	Frames	Use in The Last Ferry
Wide / establishing	Whole location, figures small	S1 market, S4 pier
Medium	Waist up	S2 Mei running
Close-up	Face or object fills the frame	S3 the red box
Two-shot / over-the-shoulder	Two characters, one favoured	S5 the hand-over
Low angle	Camera below eye line: stature, emotion	S6 Mei reads the tag
High angle	Camera above: vulnerability, detail	S3 box in her hands

Movement, Light and Composition

1

CAMERA MOVEMENT

Locked-off, pan, tilt, push-in, tracking, crane, handheld — one move per shot.

2

SPEED

Real-time or slow motion (speed ramp) for a single beat.

3

KEY, FILL, RIM

Key sets the mood, fill controls contrast, rim separates the subject.

4

PRACTICALS

Visible light sources — lanterns, the sodium lamp — motivate the grade.

5

RULE OF THIRDS

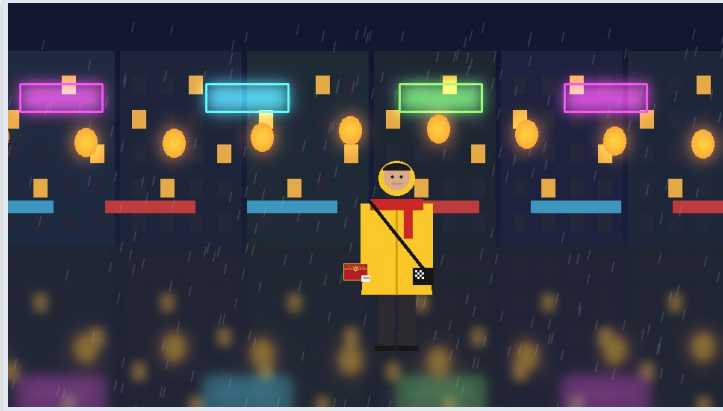
Eyes and the prop on third lines; keep the centre third for 9:16.

6

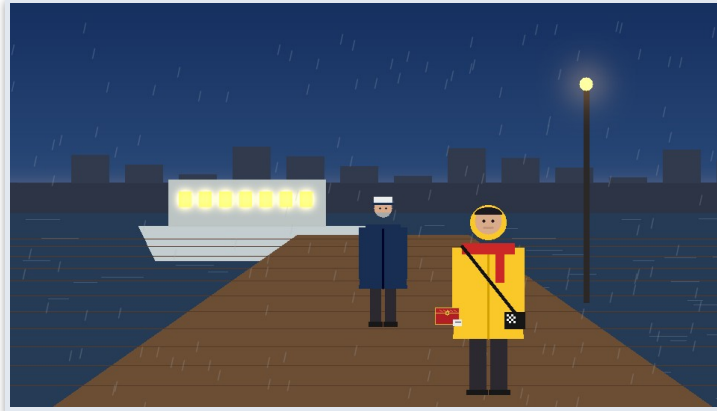
180-DEGREE RULE

Keep the camera on one side of the action so directions read.

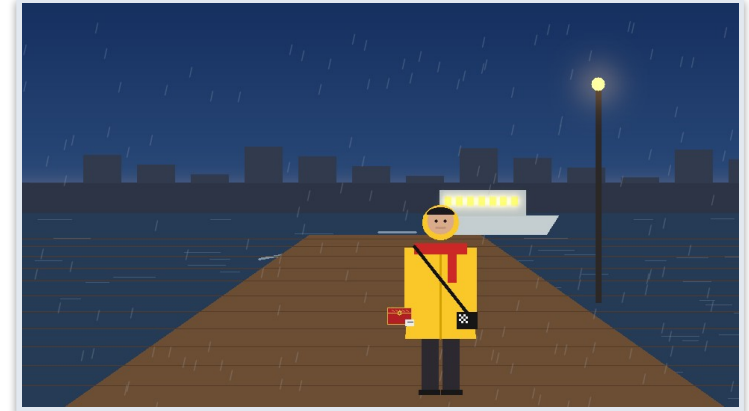
Locations Need Angles and Story States



NIGHT MARKET plate



HARBOUR PIER — state A: ferry docked



HARBOUR PIER — state B: ferry leaving

Generate state B by EDITING state A ('change nothing else') and verify it (Lab 04) — otherwise the model invents a different pier.

Lab 06 — Character Reference Sheets and Consistency Features

INPUTS

MEI reference sheet: front, side, back, close-up and the RED BOX (SIMULATED)

TECHNICAL QUESTION

Which local features catch a changed scarf, a drifted coat and a missing clasp — and which miss them?

EVIDENCE TO PRODUCE

A feature-extraction design (which feature detects which failure), an identity report for six shots with pass/drift verdicts and reasons, and an identity strip comparing each shot crop with the reference sheet.

LAB 06 · 30 MIN · Python 3 · opencv-python · NumPy · optional: an image model with character references
Detailed procedures and prompts: [Learner Guide / labs/lab-06-character-consistency-features/README.md](#)

Lab 07 — Look Bible and Scene Continuity with Global Descriptors

INPUTS

Approved look master, NIGHT MARKET (SIMULATED)

TECHNICAL QUESTION

Which global descriptor identifies each off-look shot, and why must shots be compared with their own location?

EVIDENCE TO PRODUCE

A look report for eight shots with the descriptor that moved for every off-look shot, a colour-coded look strip, and a look bible with the limits you would ship.

LAB 07 · 30 MIN · Python 3 · opencv-python · NumPy · no network

Detailed procedures and prompts: [Learner Guide / labs/lab-07-look-bible-scene-continuity/README.md](#)

TOPIC 04

AI Video Generation, Animation and Video Analytics

Deep learning for video — temporal coherence, start/end frames and video references · multi-shot and timeline prompting with dialogue and emotion · detecting and verifying the cast · voiceover-first sound design, captions and export · shot, motion, tracking and drift analytics

Key Concepts — Topic 04

1

Deep video models learn motion from data

Convolution and attention layers learn spatial structure; temporal attention links frames. Each frame must look right (spatial coherence) AND agree with its neighbours (temporal coherence): flicker, morphing and identity drift are temporal failures.

2

Continuity is conditioned, shot to shot

A start frame fixes where a shot begins; an end frame fixes where it lands. A video reference or extension passes the whole previous clip, so lighting, mood and motion carry across the cut. Clips are typically capped at 5-15 s.

3

Write shots, not paragraphs

Multi-shot prompts list Shot 1, Shot 2, Shot 3 in order; timeline prompts give second-by-second beats. Dialogue goes in quotation marks, and each character gets an emotion and an intensity. Dense action: generate three takes and pick.

4

Detection and segmentation turn frames into evidence

A face or person detector counts the cast in every shot; a segmentation mask isolates the subject for grading. Evaluate detectors with IoU, precision and recall before you trust them to approve a shot.

5

Video analytics finds cuts, motion, drift and events

Histogram differences find shot boundaries; optical flow classifies camera movement; CAMShift tracks the courier's raincoat; hue drift and luminance flicker become numbers; the first frame the subject appears is an event.

6

Sound drives picture

Generate the voiceover first and cut the storyboard to its timing. Direct delivery with audio tags, score music by mood and tempo, add sound effects, then caption inside the safe area and export one master per platform.

Spatial Coherence versus Temporal Coherence

Within a frame

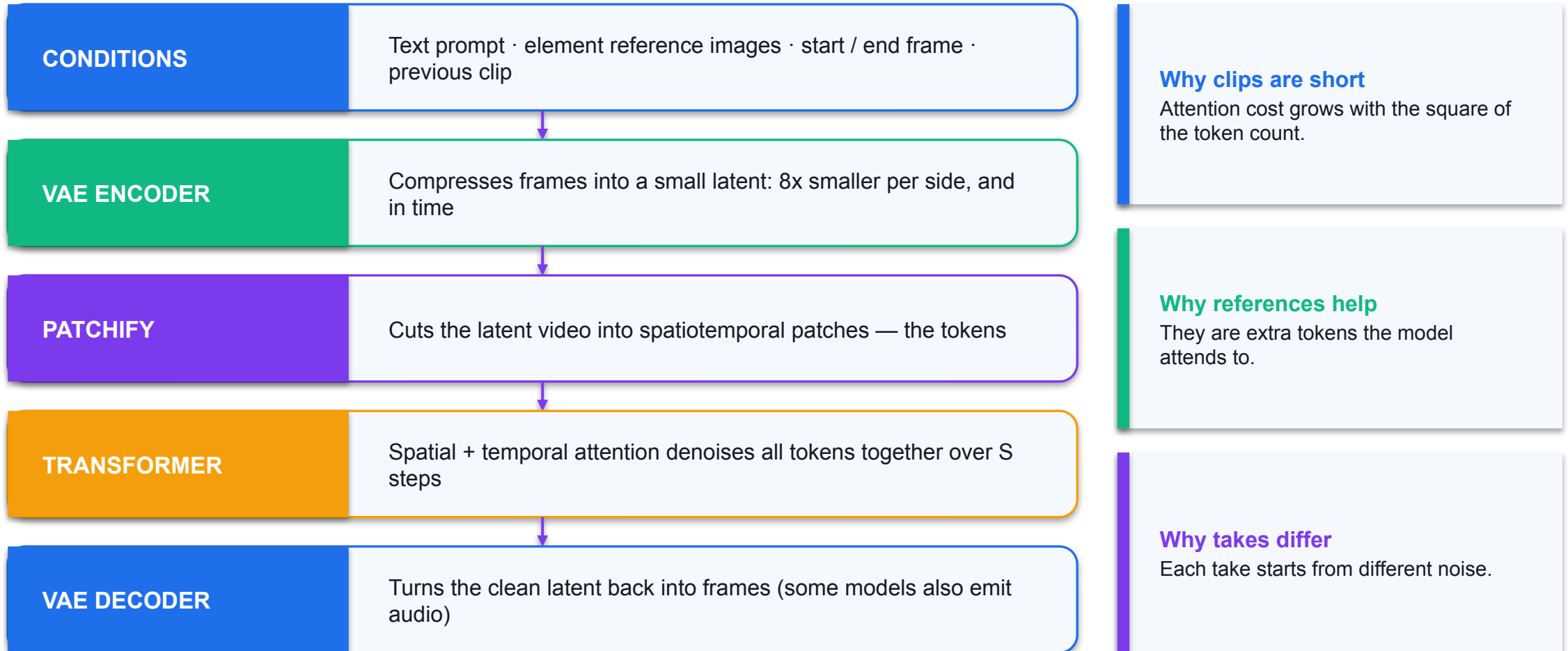
- **Spatial: is each frame right?**
 - Anatomy, perspective, text, lighting in one frame.
 - Checked frame by frame.
 - Image models have mostly solved this.

Across frames

- **Temporal: do the frames agree?**
 - Identity, wardrobe and background stay put.
 - Motion obeys physics; no flicker, no morphing.
 - A per-frame score cannot see it.
 - The defining problem of AI video.

Lab 10 measures the temporal failures — flicker and identity drift — that a frame-by-frame review misses.

Inside a Video Diffusion Transformer



Why Clips Are Capped at a Few Seconds

Tokens

$$N = (F/f) \times (H/p) \times (W/p)$$

F frames, H x W latent, compressed by f in time and p in space.

Attention

$$\text{cost} \sim N^2$$

Doubling the duration roughly quadruples full attention cost.

Credits

$$\text{generated } s = \text{sum}(\text{seconds} \times \text{takes})$$

The Last Ferry: 48 s finished, about 98 s generated with extra takes.

Typical caps today are 4–15 seconds per clip. Longer scenes are built from shots joined by start/end frames or video extension.

Continuity Controls — What Each One Fixes

Control	You give the model	It fixes	Limitation
Element references	Saved character, location, prop images	Who and where	Not the motion
Start frame	The graded still the shot begins on	Composition, look, placement	The end can wander
Start + end frame	Where the shot begins and lands	The whole path of the shot	Mood can reset at the cut
Video reference / extension	The entire previous clip	Light, mood, motion across the cut	Errors carry over too
Takes	Several generations of one shot	Luck: pick the best performance	Multiplies credits

Director Controls in a Video Generator

1

GENRE AND STYLE

Action, drama, horror — sets pacing and visual register.

2

CAMERA MOVEMENT

Choose one move, or let the model pick for a single beat.

3

SPEED RAMP

Slow motion on one moment, real time elsewhere.

4

DURATION, RESOLUTION, ASPECT

Draft short and small; match the delivery ratio.

5

EMOTION AND INTENSITY

Per character: vigilance, rage, trust — low to high.

6

AUDIO ON / OFF

Native dialogue and ambience, or silent for your own mix.

A Multi-Shot Prompt with Dialogue and Emotion

```
@mei @captain_tan @harbour_pier_docked @red_box
Duration 6 s, 16:9, medium two-shot, over-the-shoulder, static.

Shot 1: MEI, out of breath, holds out the red box.
Shot 2: CAPTAIN TAN (warm, intensity medium) does not take it;
        he nods at the folded tag.
        CAPTAIN TAN: "It's not for me. Read the tag."
Shot 3: close on MEI's face (curious, intensity low).

Look: blue hour, teal shadows, warm sodium lamp, light rain.
Constraints: coat and scarf exactly as @mei; no extra people.
```

Shots in order

One line per shot so the model need not guess timing.

Dialogue in quotes

Read as a voice line for that character.

Emotion per character

With an intensity, like directing a cast.

Constraints last

What must not change, stated plainly.

A Timeline Prompt, Second by Second

```
SHOT S1 (8 s, 16:9) – wide, eye level, slow pan right, 24mm  
[0.0s] Lanterns sway; the camera finds MEI in the crowd.  
[1.0s] MEI (V.O.) (breathless, medium):  
        "Last ferry leaves at midnight. Four minutes."  
[5.0s] She checks her watch without slowing.
```

```
Start frame: reference/board/shot1_start.png  
End frame:   reference/board/shot1_end.png  
Takes: 1     Negative: extras in foreground, text, daylight
```

Beats have times

The voice line lands where the VO was recorded.

Start and end frames

Fix where the shot begins and lands.

Takes by density

S2 is dense action: three takes.

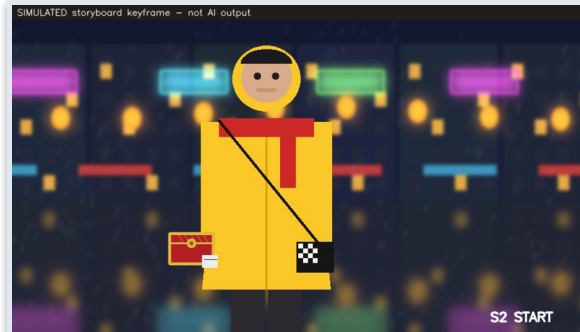
Three formats

Prose, timeline and JSON for every shot.

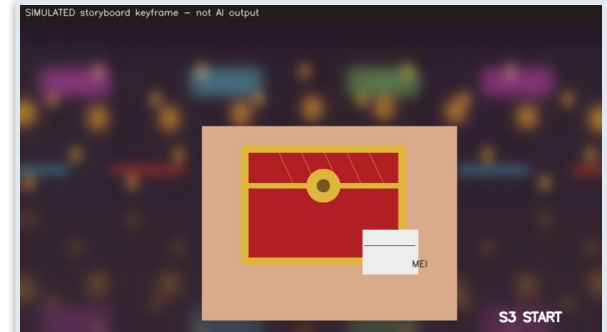
The Storyboard: Start Frames of Six Shots



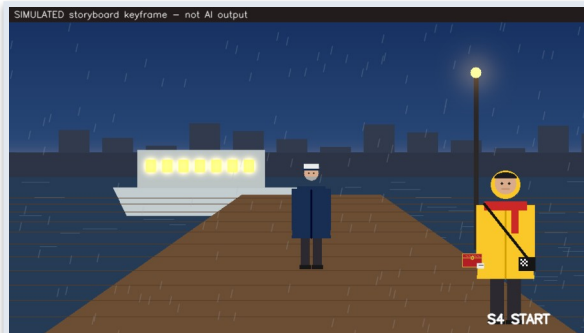
S1 wide · 8 s



S2 medium · 8 s



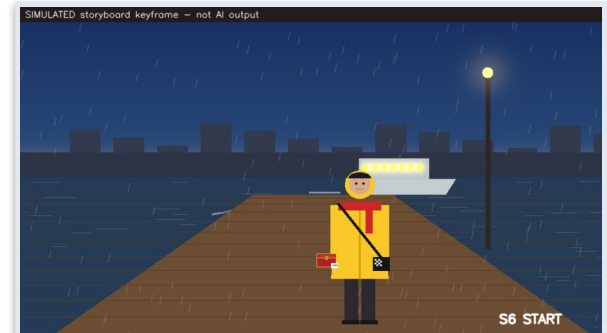
S3 close-up · 6 s



S4 wide · 8 s



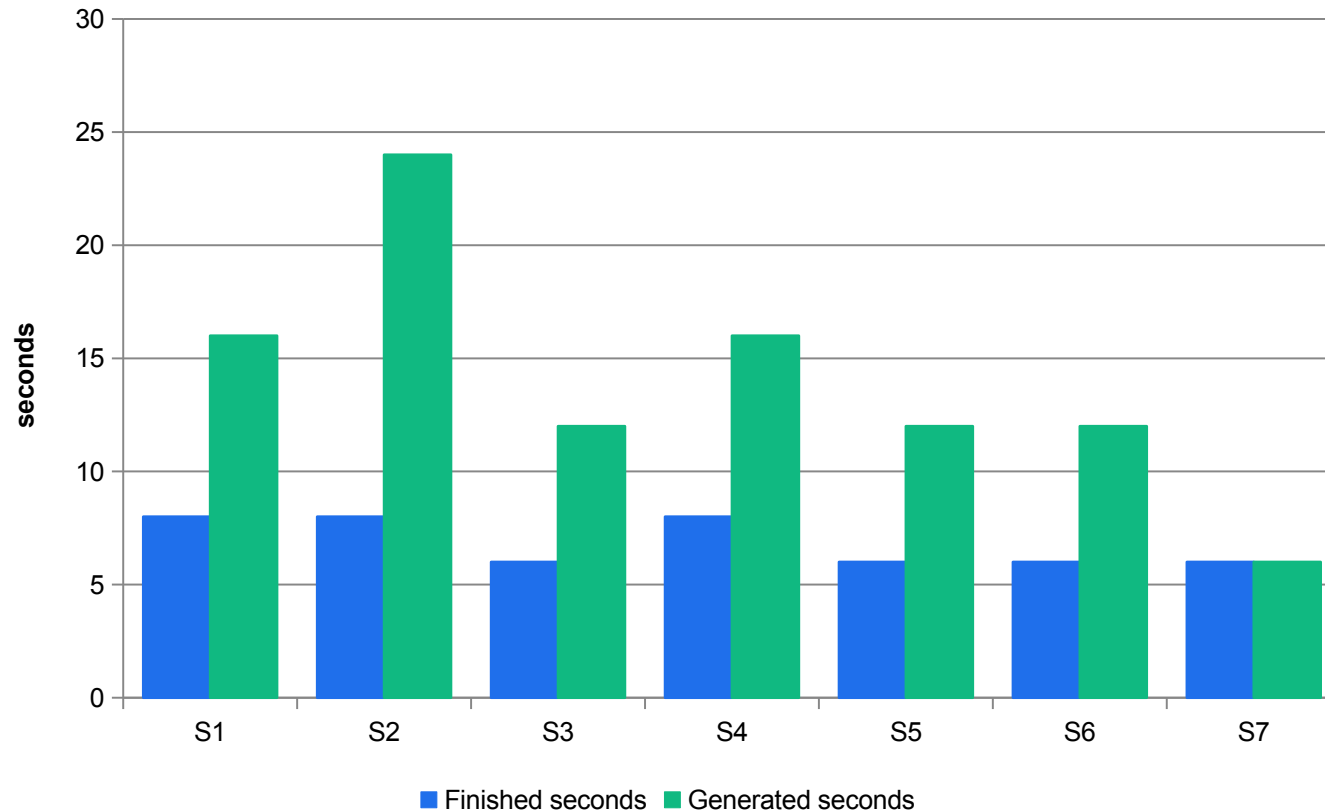
S5 two-shot · 6 s



S6 low angle · 12 s

The check finds two problems: S6 exceeds the 10-second clip cap (split it into S6 + an extension S7) and one voice line crosses a cut.

Seconds Generated per Shot, Including Takes



2x overhead

Extra takes double the generated seconds.

Dense action

S2 gets three takes; S7 extends S6 with one.

Draft first

Generate low-resolution drafts, upscale keepers.

Plan the takes

Takes are a budget line, not an accident.

MODELLED

From the Lab 12 production plan: seconds x takes per shot. 48 finished seconds cost 98 generated seconds.

Classification, Detection, Segmentation

Classification

- One label per image.
- 'Is this shot at night?'
- Metric: accuracy, F1.
- Cheapest; least detail.

Detection

- A box + label + score per object.
- 'How many faces are on screen?'
- Metric: IoU, precision, recall.
- Lab 08: cast presence.

Segmentation

- A label per pixel.
- 'Which pixels are Mei's raincoat?'
- Metric: mask IoU.
- Labs 10–11: tracking, reframing.

Evaluating a Detector Before You Trust It

IoU

$$|A \cap B| / |A \cup B|$$

$[0,0,100,100]$ vs $[50,0,100,100]$ = $5000 / 15000 = 0.333$.

Match

same class, IoU $\geq t$,
one-to-one, best score first

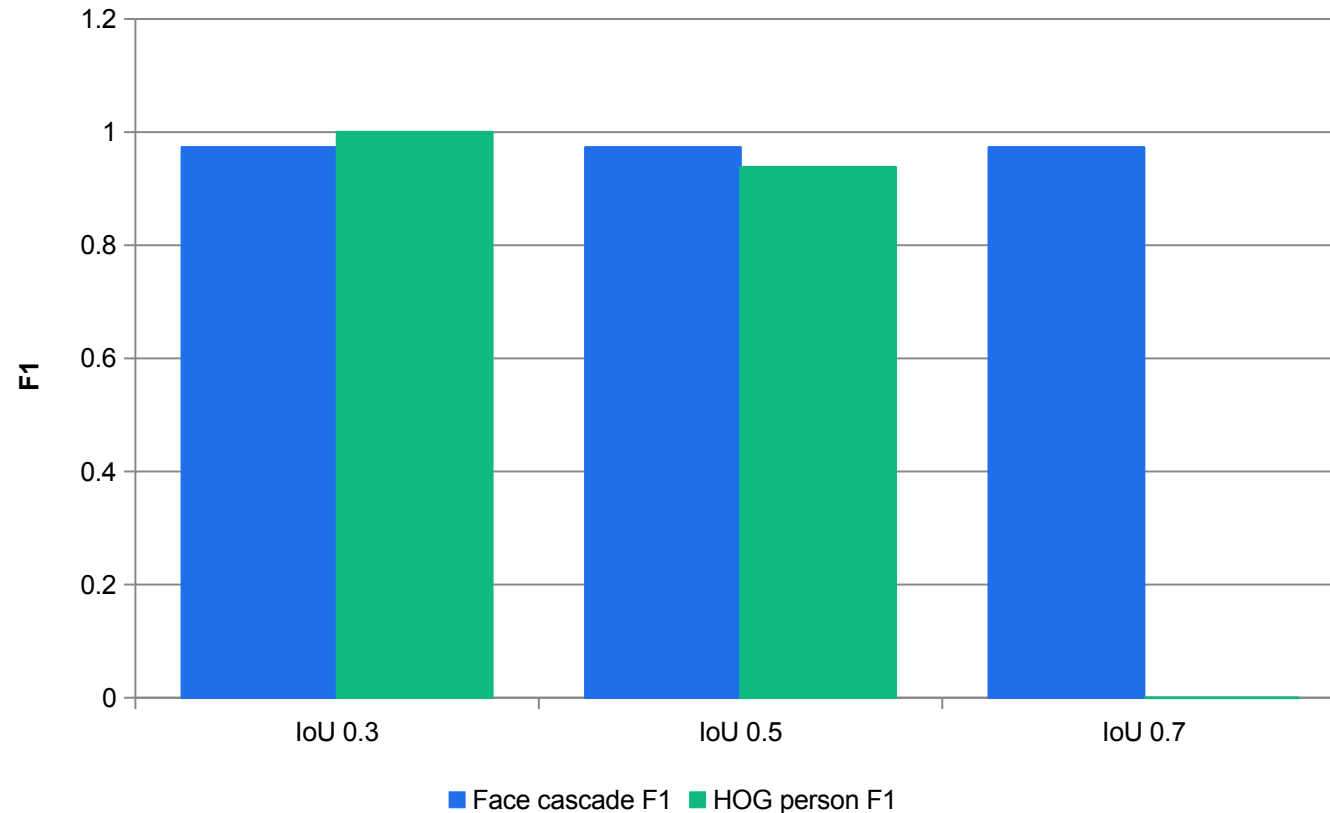
Unmatched predictions are FP;
unmatched truths are FN.

Precision / recall

$$\begin{aligned} P &= TP / (TP + FP) \\ R &= TP / (TP + FN) \\ F1 &= 2PR / (P + R) \end{aligned}$$

State the IoU threshold with every number.

Two Built-In Detectors at Three IoU Thresholds



Face cascade

One false positive; tight boxes at every IoU.

HOG people

Right objects, loose box edges: collapses at 0.7.

Positive control

On an AI-generated photo-real still, many false positives.

Domain shift

Evaluate on your own footage before automating.

MEASURED

detect_eval.py on 12 SIMULATED schematic storyboard frames with exact truth. Does not transfer to photographic footage.

Cast Check: Is the Frame Wrong, or the Detector?

Frame	Shot list expects	Detector sees	Truth	Verdict
frame_04	2 faces	1	1	CAST DRIFT — a character is missing
frame_06	1 face	2	1	DETECTOR ERROR — false positive
frame_09	1 face	2	2	CAST DRIFT — an extra appeared
9 others	as listed	as listed	as listed	OK

Only a disagreement between the shot list and the detector is routed to a reviewer; the truth column is what the reviewer decides.

Lab 08 — Cast-Presence Detection and Detector Evaluation

INPUTS

Twelve schematic storyboard frames with exact truth (SIMULATED)

TECHNICAL QUESTION

How do confidence and IoU thresholds change false positives and misses — and is a flagged frame wrong, or is the detector?

EVIDENCE TO PRODUCE

An evaluation table (TP, FP, FN, precision, recall, F1 at three IoU thresholds), a parameter sweep with a chosen operating point, a cast-check table separating CAST DRIFT from DETECTOR ERROR, and a positive-control run on an AI-generated still.

LAB 08 · 60 MIN · Python 3 · opencv-python · NumPy · no weights download, no network

Detailed procedures and prompts: [Learner Guide / labs/lab-08-cast-detection-evaluation/README.md](#)

Lab 09 — Storyboard to Shot Prompts, Start and End Frames, and the Animatic

INPUTS

Storyboard start and end keyframes for six shots (SIMULATED)

TECHNICAL QUESTION

Do the shot durations, clip-length cap and voiceover timing agree — before any credit is spent?

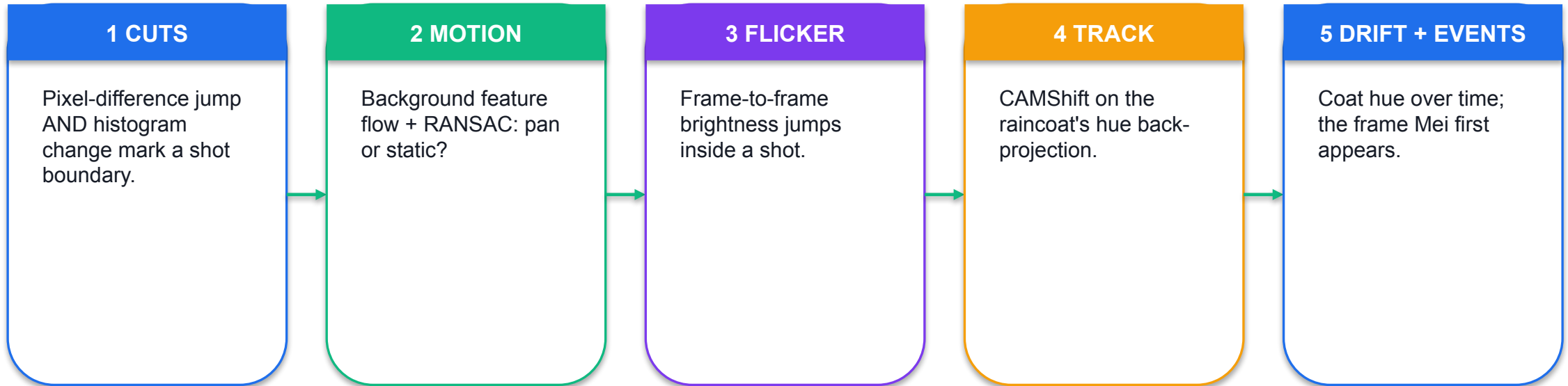
EVIDENCE TO PRODUCE

A corrected shot list that passes every check, a shot-prompt pack (prose, timeline and JSON per shot, with start/end frames and takes), and a 48-second animatic with the voiceover lines burned in.

LAB 09 · 30 MIN · Python 3 · opencv-python · NumPy · optional: any image-to-video model with start/end frames

Detailed procedures and prompts: [Learner Guide / labs/lab-09-storyboard-shot-prompts/README.md](#)

Video Analytics on the Assembled Sequence



SO WHAT Each measure answers one editorial question — where are the cuts, what does the camera do, which shot flickers, who drifts, and when does she enter — with a number and a timecode.

Three Analytics Rules

Cut

```
diff_t > max(12, 5 x median(diff_recent))  
AND hist_dist_t > 0.08
```

A flicker passes the first test but not the second: brightness changes, palette does not.

Flicker index

```
mean | L_t - L_(t-1) |  
L = mean frame brightness
```

Above 4 inside one shot = a flickering generation.

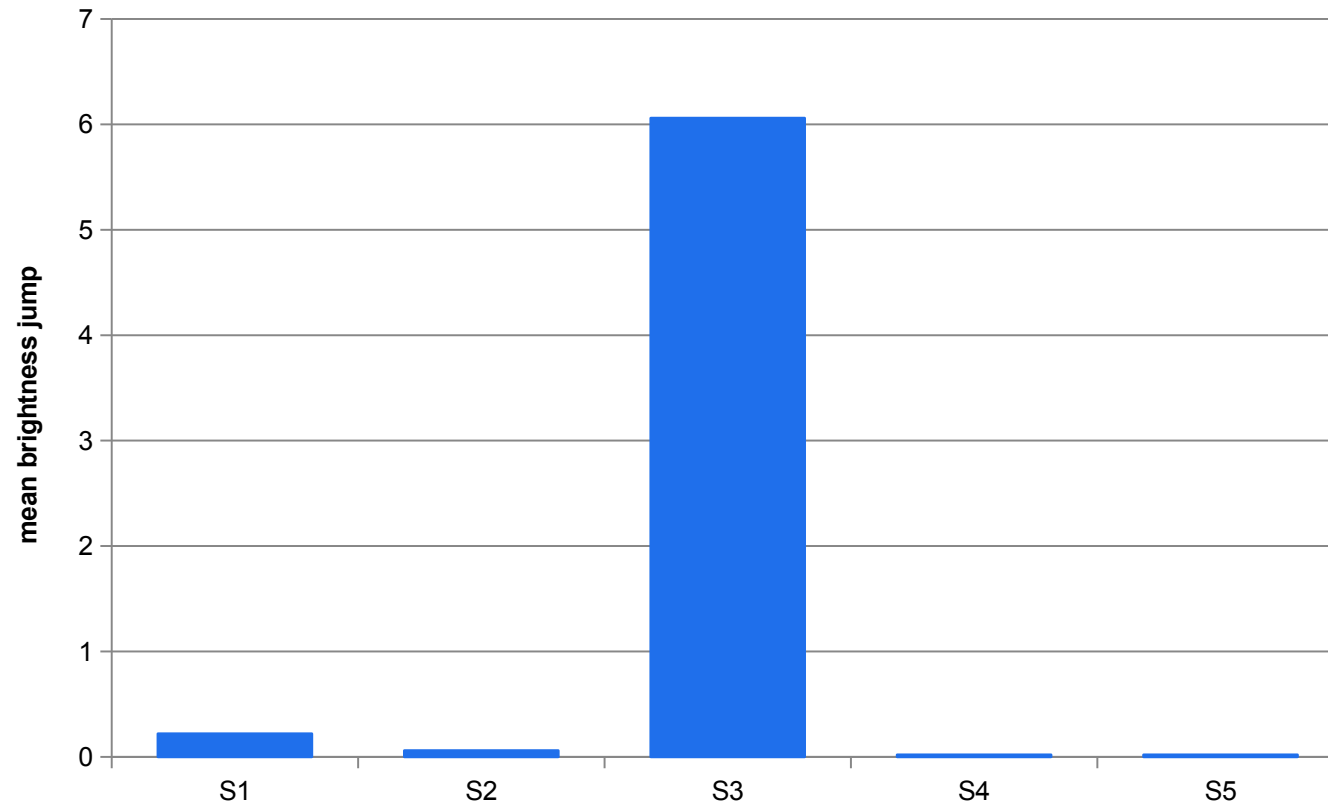
Identity drift

```
hue_end - hue_start  
of the tracked coat
```

More than 4 hue units within one shot = the character changed.

Every threshold is a design decision: report the value and the evidence for it.

Flicker Index per Shot



MEASURED

continuity.py on the SIMULATED 16-second sequence with an injected flicker in S3. Threshold 4.0.

Cuts: 4 of 4

Frames 72, 168, 216, 312 — no false cuts.

S1 pans right

Background moves about -5 px per frame.

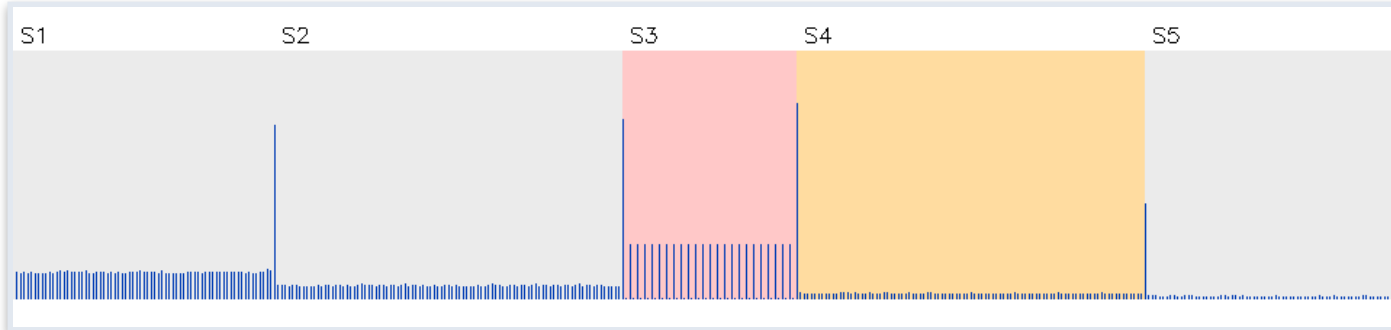
S3 flickers

6.06 against a threshold of 4.

Mei enters at 3.00 s

Frame 72 — the event for the edit list.

The Continuity Timeline



Blue spikes = frame differences (cuts) · red = flicker shot · orange = drift shot

S2 tracked cleanly

IoU(x) 0.82, jitter 1 px, coat hue steady at 23.

S4 drifts

Coat hue 23 → 13 across the shot.

Drift breaks tracking

The colour tracker loses her as the hue leaves.

Fix

Regenerate S4 with @mei re-attached; video-reference S3.

Tracking the Subject

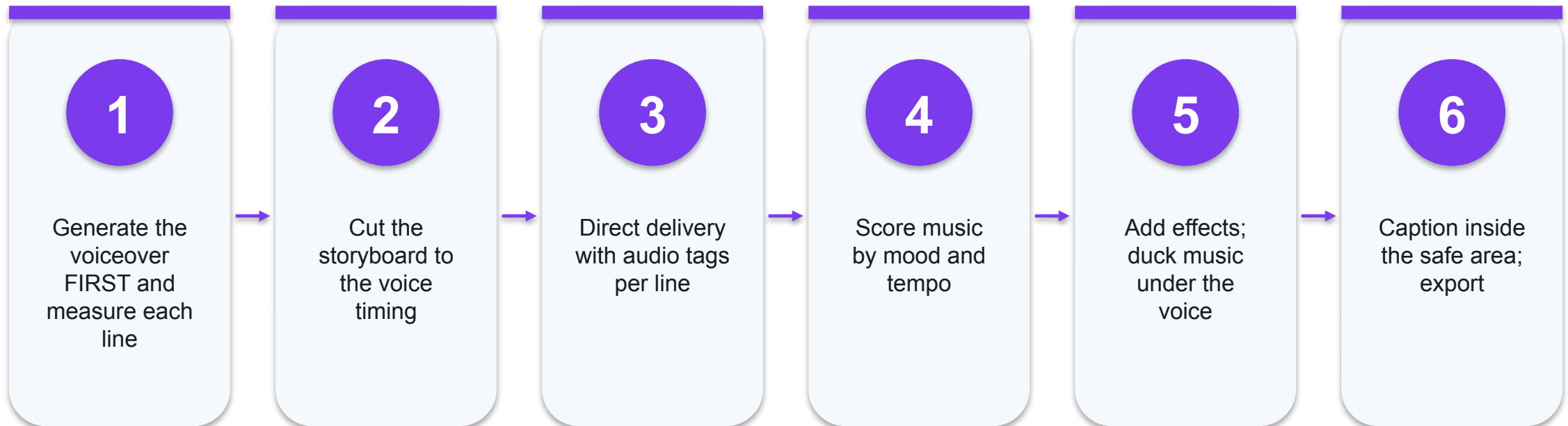
Mechanism

- **How CAMShift follows MEI**
 - Build a hue histogram of the raincoat in the first box.
 - Back-project it: each pixel scores 'how raincoat-like'.
 - Climb to the densest window; adapt its size and angle.
 - Repeat every frame — no training data needed.

Use

- **Why it matters for the film**
 - Subject-following crops for the 9:16 cut-down.
 - Jitter and scale drift expose unstable generations.
 - A colour tracker fails when the colour drifts.
 - That failure is itself evidence of identity drift.

Sound Drives Picture

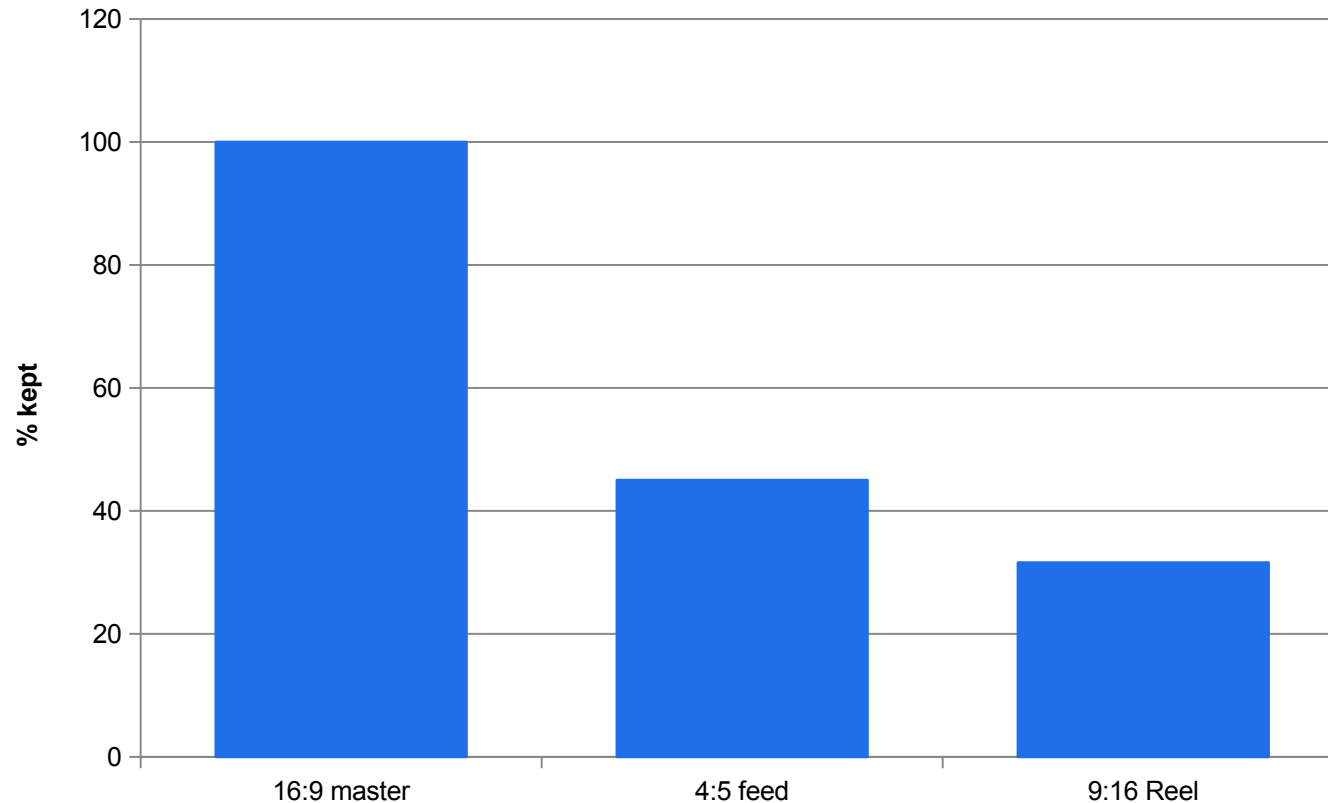


Keep one voice ID per character across every clip, or the voice changes between shots even when the face does not.

Caption Rules the Export Checks

Rule	Limit	Why
Lines per caption	2	More covers the picture
Characters per line	42	Readable on a phone
Reading speed	17 characters / second	Viewers must finish the line
On screen	1.0 – 7.0 seconds	Too short flashes; too long lingers
Gap between captions	≥ 0.08 seconds	Separate lines read as separate lines
Position	Inside the platform safe area	Buttons and handles cover the rest

The Export Ladder



MEASURED

finish.py: 240 frames per export; with FFmpeg each is muxed to H.264 video + AAC audio and verified with ffprobe.

Subject-following crop

The 9:16 and 4:5 crops follow Mei's raincoat mask.

Mix

Peak -1.5 dBFS; voice 13 dB over the ducked music.

Label travels

The SIMULATED banner is re-burned on every derivative.

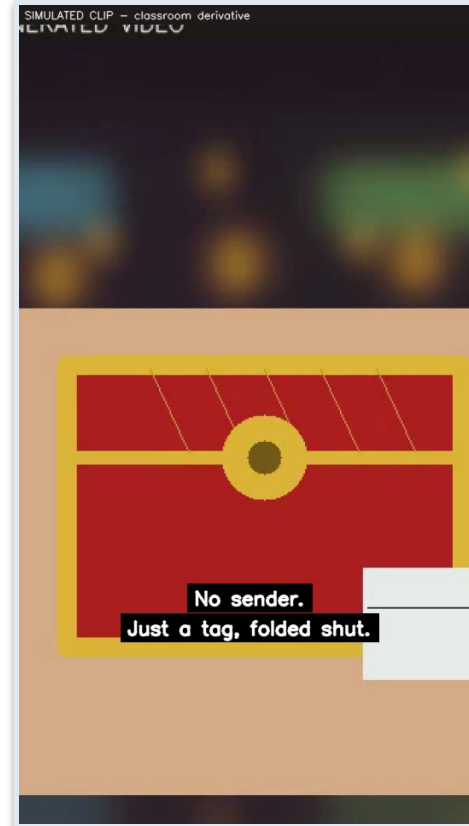
Disclosure

Set the platform's AI-content label on upload.

Finished Frames: Master and Reel



16:9 master with captions



9:16 Reel: subject-following crop, captions above the UI zone

Lab 10 — Shot, Motion and Continuity Analytics on the Assembled Sequence

INPUTS

16-second, five-shot sequence with injected flicker and identity drift (SIMULATED clip)

TECHNICAL QUESTION

Do the cut detector, camera-motion estimate, flicker index, tracker and hue-drift measure tell the same story as the truth file?

EVIDENCE TO PRODUCE

A shot report (boundaries, camera movement, flicker), a tracking report (IoU, centre error, jitter, hue drift), the first-appearance event, and a timeline graphic, each compared with the truth file.

LAB 10 · 60 MIN · Python 3 · opencv-python · NumPy · no network

Detailed procedures and prompts: [Learner Guide / labs/lab-10-shot-motion-continuity-analytics/README.md](#)

Lab 11 — Voiceover, Music, Captions and the Final Export Ladder

INPUTS

10-second picture lock of the last three shots (SIMULATED clip)

TECHNICAL QUESTION

Do every caption and the mix meet the delivery rules, and does each export keep the subject in frame?

EVIDENCE TO PRODUCE

A caption file that passes every rule, an audio report (peak, RMS, voice-over-music margin), and three exported masters with burned-in captions — muxed to H.264/AAC when FFmpeg is installed.

LAB 11 · 30 MIN · Python 3 · opencv-python · NumPy · optional FFmpeg · optional: a voice, music and editing tool

Detailed procedures and prompts: [Learner Guide / labs/lab-11-sound-captions-final-export/README.md](#)

TOPIC 05

Cloud and Edge AI Workflows for Short Film Production

The four-layer production architecture · all-in-one agents, multi-tool cloud pipelines and local open models · the protocols that move prompts, references and renders · cost, latency, privacy, likeness and disclosure constraints · measured cloud-versus-local trade-offs

Key Concepts — Topic 05

1

A production system has four layers

Assets (script, references, voice, footage) -> local/edge workstation (pre-processing, QA, edit, grade, optional local models) -> cloud generation (image, video, voice and music services) -> delivery (platforms, archive).

2

Three workflow patterns, three control levels

An all-in-one agent turns one prompt and one photo into a finished drama. A multi-tool cloud pipeline chains planner, image, video, voice, music and editor. A local or hybrid pipeline runs open models on your own GPU.

3

Protocols are chosen by payload and wait time

HTTPS + JSON to submit a job; a long-running operation you poll, or a webhook that calls you back; WebSocket progress events from a local node server; signed-URL uploads to object storage for references and renders; HLS for streaming delivery.

4

Cost and time scale with shots x takes x seconds x resolution

Cost per finished minute = total credits spent / minutes that survive the edit. Explore at low resolution, generate extra takes only for dense action, and upscale only the keepers.

5

Constraints are gates, not weights

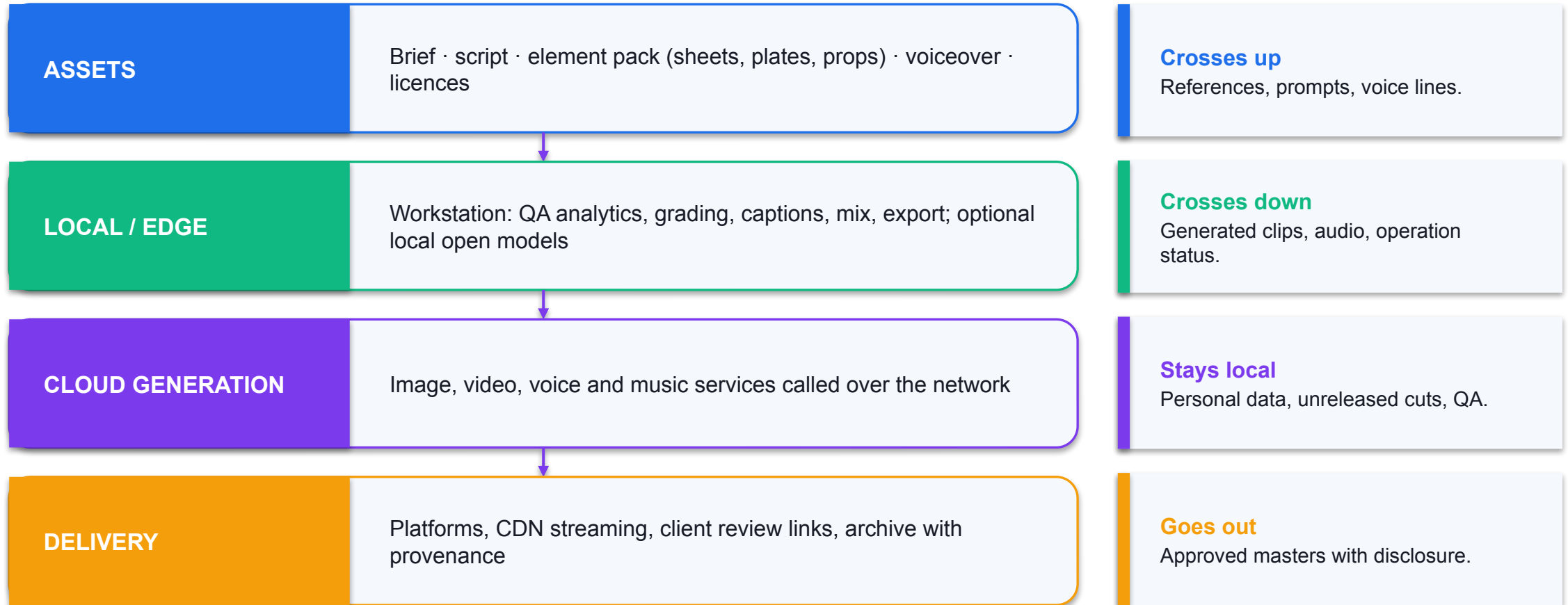
Consent for any real face or voice, licence terms for commercial use, personal-data rules (PDPA), platform AI-content disclosure, provenance (C2PA, SynthID) and brand safety eliminate options before cost is argued.

6

Local compute is bounded by memory

Weight memory = parameters x bytes per parameter; quantising from 32-bit to b-bit scales it by $b/32$. Local keeps data private and has no per-clip fee; cloud is faster to start, higher quality and needs no GPU.

A Short-Film Production System Has Four Layers

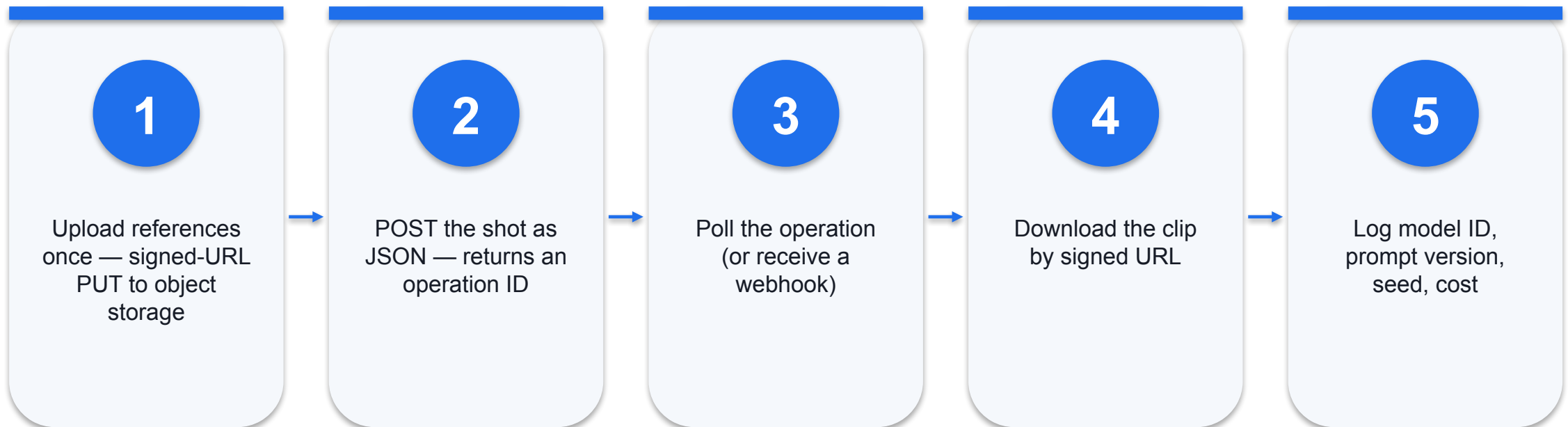


The design decision is the split: which stages run locally and which in the cloud — and what data crosses each boundary.

Three Workflow Architectures Compared

	All-in-one agent	Multi-tool cloud pipeline	Hybrid / local
Control per shot	Low	High	High
Speed to first draft	Minutes	Hours	Hours
Consistency tools	Agent's own	Elements, start/end frames, extension	Same + local QA
Data leaves premises	Yes	Yes	Only what you send
Skills needed	Prompting	Prompting + editing	Plus local set-up
Best for	Social drafts, volume	Brand hero films	Series, privacy, cost control

How a Cloud Video Job Actually Travels



Generation takes seconds to minutes, so APIs are asynchronous: a long-running operation you check, not a request you hold open.

Protocols Chosen by Payload and Wait Time

Protocol	Carries	Where in the film pipeline
HTTPS + REST/JSON	Prompts, parameters, job status	Submitting every shot and still
Long-running operation + polling	An ID you check until done	Video generation (seconds–minutes)
Webhook (HTTP callback)	A 'job finished' event pushed to you	Batch generation at scale
WebSocket	Streaming progress events	Local node servers; live previews
Signed URL to object storage	Large files: references, renders	Upload element pack, download clips
HLS / RTMP	Segmented or live video streams	Delivery and review screenings
C2PA manifest (in the file)	Provenance: who, what tool, what edits	Travels with every master

The Asynchronous Pattern in Code

```
operation = client.models.generate_videos(  
    model=VIDEO_MODEL_ID,          # record the exact ID and date  
    prompt=shot['timeline'],  
    image=start_frame,             # image-to-video  
    config={'aspect_ratio': '16:9', 'duration_seconds': 8})  
  
while not operation.done:          # long-running operation  
    time.sleep(10)  
    operation = client.operations.get(operation)  
  
clip = operation.response.generated_videos[0]  
client.files.download(file=clip.video)
```

Submit

Returns immediately with an operation handle.

Poll

Check every few seconds; back off politely.

Download

The result is a file reference, not the bytes.

Same shape everywhere

Most video APIs follow this submit-poll-fetch pattern.

VERIFY Parameter names differ by provider; check the current API reference before use.

Modelling Cost, Time and Upload

Cost

```
S$ = sum(sec x takes) x credits/s  
    x S$/credit + fixed
```

The Last Ferry: 98 generated seconds for 48 finished seconds.

Cost per finished minute

```
S$ / (finished_s / 60)
```

The number to compare with a live shoot or an agency quote.

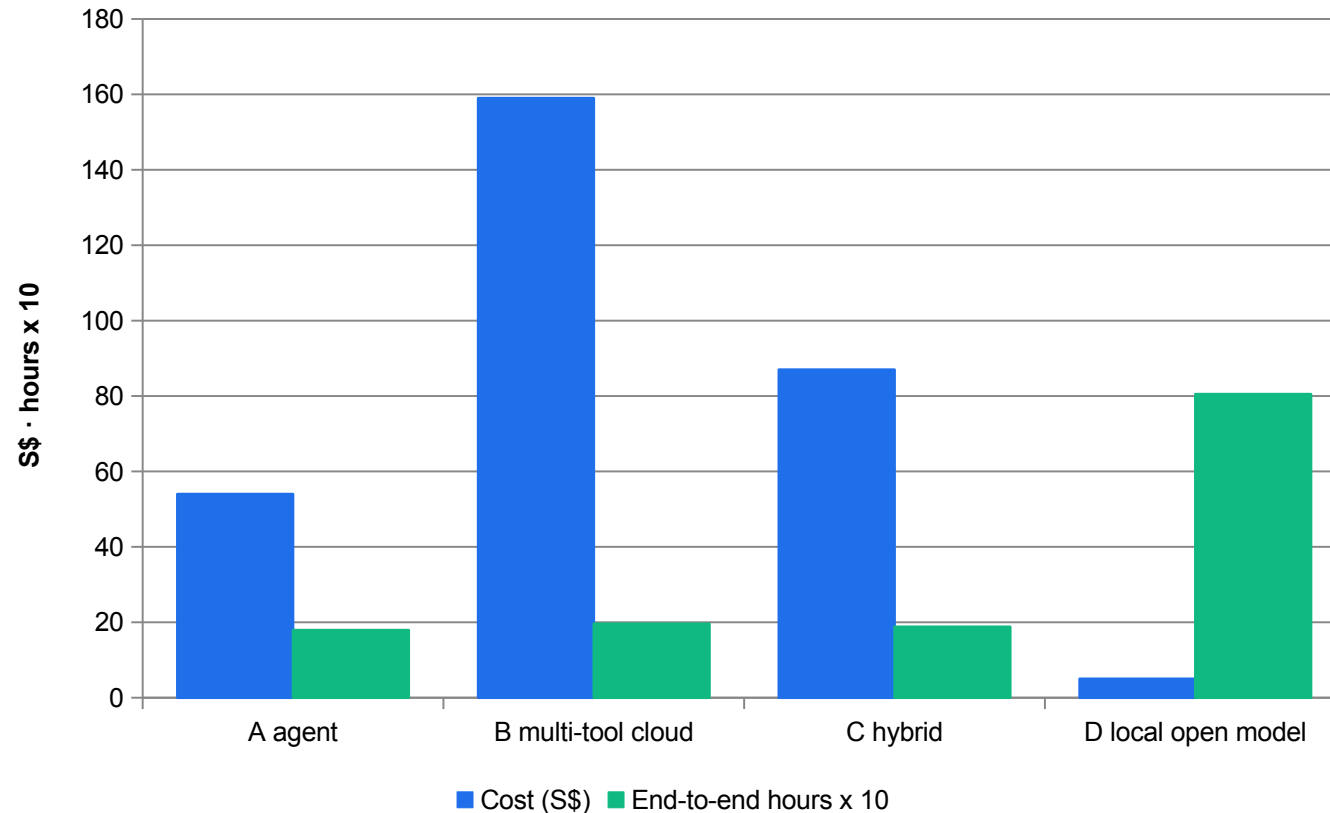
Upload

```
t = bytes x 8 / uplink_bps
```

0.44 MB element pack at 20 Mb/s = 0.2 s: references are cheap to send.

Lab 12 MEASURES the local stages on your own laptop and MODELS the cloud stages from the production plan.

Four Architectures for The Last Ferry



A fails two gates

Quality tier 3 < 4; no per-shot control.

D fails quality

Cheapest, but tier 2 for a brand hero film.

B and C pass

C is cheaper: S\$87 vs S\$159.

Recommend C

Cloud hero shots + local QA, grade and export.

ILLUSTRATIVE

Modelled from ILLUSTRATIVE credit rates and render speeds in labs/lab-12/data/architectures.json — replace with current prices.

Will an Open Video Model Fit the GPU?

Weights

```
bytes = params x bits / 8
```

14 B parameters: 28 GB at FP16, 14 GB at INT8, 7 GB at INT4.

With activations

```
x 1.3 overhead
```

36.4 GB FP16 · 18.2 GB INT8 · 9.1 GB INT4.

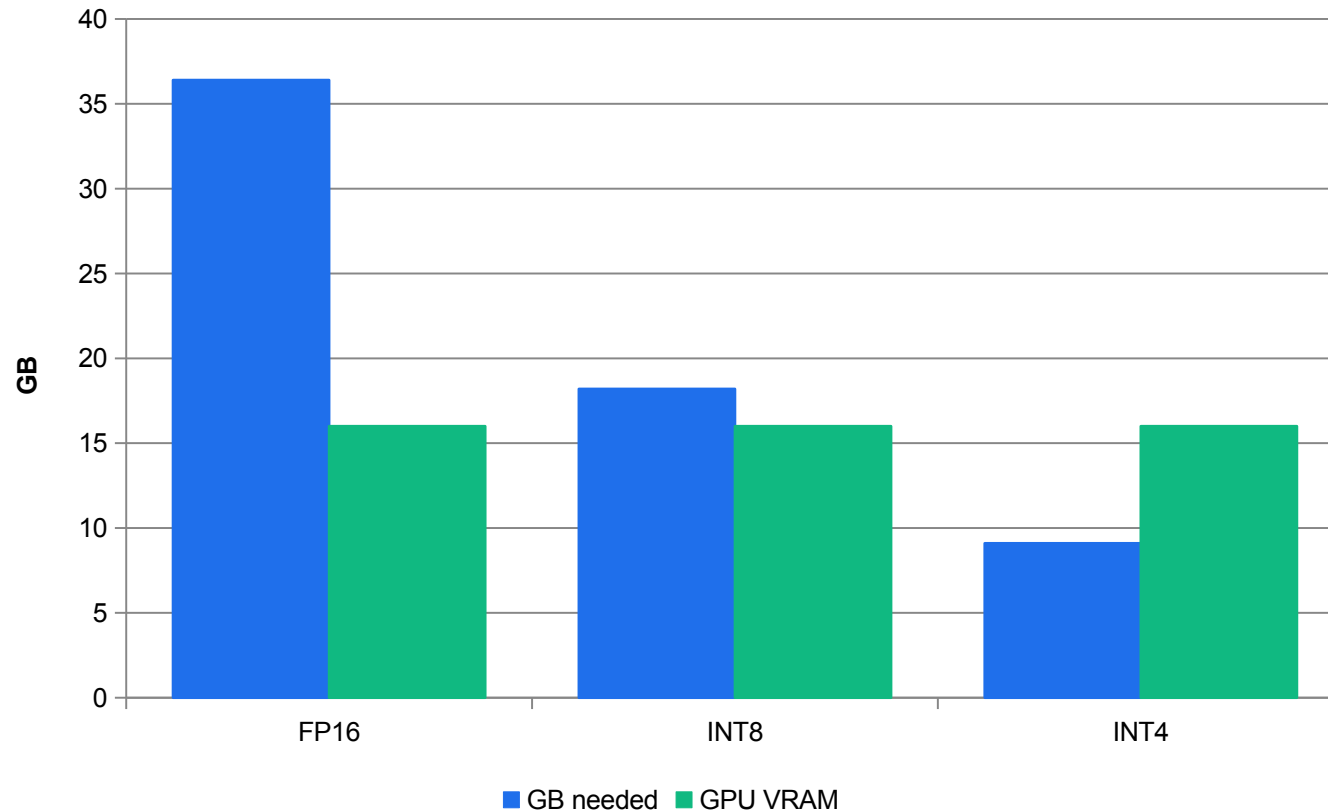
Decision

```
fits if need <= VRAM
```

On a 16 GB GPU only INT4 fits — at a quality cost.

Quantising from 32-bit to b-bit scales weight memory by $b/32$. Offloading trades memory for time.

Memory Needed by Precision (14 B-Parameter Model)



FP16

More than twice the GPU.

INT8

Still does not fit 16 GB.

INT4

Fits, with quality to verify.

Or rent

A cloud GPU is a hybrid option.

MODELLED

params x bytes x 1.3 activation overhead, as computed by production_arch.py. Real usage varies by model and resolution.

Real-World Constraints Are Gates

1

CONSENT AND LIKENESS

A real face or voice without written consent removes the option — whatever it costs.

2

PERSONAL DATA (PDPA)

Customer and staff photos may not leave approved systems or regions.

3

LICENCE TERMS

Commercial use, training on uploads, ownership of outputs.

4

QUALITY AND CONTROL

Brand hero work needs per-shot control and a quality tier.

5

DISCLOSURE AND PROVENANCE

Platform AI labels, C2PA credentials, invisible watermarks.

6

BUDGET AND DEADLINE

Only then compare cost and time among what survives.

What-If: One Change, New Decision

Change	Survivors	Recommendation
Baseline brief	B, C	C hybrid — cheapest that passes
Likeness data must stay local	D only (fails quality)	None: change the plan, e.g. fictional cast
Quality tier 2 (social draft)	B, C, D (A lacks shot control)	D local — S\$5, but 8 h
Deadline 4 hours	B, C (D takes 8 h)	C hybrid
Budget S\$80	A (fails quality), C over budget	None: cut takes or draft resolution

Never relax a gate to manufacture a winner — change the plan instead.

Cloud versus Local, Honestly

Cloud

- **Cloud strengths**

- Best available quality and newest models.
- No GPU to buy; scales instantly.
- Integrated voice, music, editing.
- Costs: per-credit spend, data leaves premises, terms change.

Local / edge

- **Local strengths**

- Data stays on your machine.
- No per-clip fee; repeatable workflows.
- Full control of models and versions.
- Costs: GPU memory, slower renders, set-up and upkeep.

Most studios land on a hybrid: generate in the cloud, keep QA, grade, captions, mix and export local.

Lab 12 — Cloud, Local and Hybrid Production Architecture Evaluation

INPUTS

4-second 960x540 clip used to time the local stages (SIMULATED clip)

TECHNICAL QUESTION

Which architecture passes every gate at the lowest cost — and what single change would flip the decision?

EVIDENCE TO PRODUCE

A four-layer architecture diagram for your chosen design, an architecture matrix (cost, cost per finished minute, generation time, upload time, end-to-end hours, VRAM fit, gates), the protocol sequence for your design, and two what-if results.

LAB 12 · 60 MIN · Python 3 · opencv-python · NumPy · a pen or diagram tool · no paid service required
Detailed procedures and prompts: [Learner Guide / labs/lab-12-cloud-local-production-architecture/README.md](#)



WRAP-UP

What You Achieved, and What Comes Next

What You Achieved

1

LO1 · Vision system concepts and applications

Scored six film ideas with gates, chose workflows and specified the production pipeline for The Last Ferry. (Lab 01)

2

LO2 · Image processing

Planned formats, linted and measured structured prompts, verified masked edits and scored grading and enhancement with PSNR and SSIM. (Labs 02–05)

3

LO3 · Feature extraction

Designed a character-consistency check from local features and audited the sequence against a numeric look bible. (Labs 06–07)

4

LO4 · Machine-learning computer vision

Evaluated two detectors, checked the cast in every frame, and wrote shot prompts with start/end frames and a timed animatic. (Labs 08–09, 11)

5

LO5 · Video analytics

Found cuts, camera motion, flicker, identity drift and the entrance event, and finished the film with captions and an export ladder. (Labs 10–11)

6

LO6 · Edge versus cloud evaluation

Measured local stages, modelled four architectures, applied the gates and defended a hybrid design. (Lab 12)

EVERY EXTERNAL CLAIM IN THIS DECK IS TRACEABLE

Source Register — Primary Sources Used (1 of 2)

Ref	Source	Locator
S1	Course Proposal Tertiary Infotech CA-WSQ-2020-013290-v2 (SSG-approved)	Part 4 Section E, Curriculum Key Features table
S2	Assessment Plan_OpenCV_TGS-2020505925_V1.0 (28 Dec 2020)	Assessment Duration table; WA and PP specification table
S3	Competency Mapping.docx (TSC ICT-DIT-4022-1.1)	Course Mapping section and ELO/A-code matrix
S4	Computer Vision Technology.docx (Skills Framework TSC extract)	TSC Proficiency Level 4 (ICT-DIT-4022-1.1) knowledge and
S5	Tertiary Courses public registration page and LMS-TMS course record	https://www.tertiarycourses.com.sg/wsqa-create-short-video
S6	Youri van Hofwegen, 'EXACTLY How to Make an AI Short Film (Full Workflow)'	Full video, 16:49
S7	Mira AI, '6 AI Video Tools I'd Use To Make An AI Short Film', YouTube, 14	Full video, 10:04
S8	Thomas Creates, 'How to Start Making AI Short Films in 2026', YouTube, 18	Video description and chapters, 12:42
S9	Roboverse, 'The Complete Guide to Making Cinematic AI Videos (2026)', YouTube	Chapters, 10:30

The full register, including exactly what each source anchors and where it is used, is reproduced in the Learner Guide.

Source Register — Primary Sources Used (2 of 2)

Ref	Source	Locator
S10	Youri van Hofwegen, 'How to Make Your First AI Movie (Full Guide)', YouTube	Video description, 17:03
S11	ElevenLabs, 'How to Make Your First AI Short Film (Full Tutorial)', YouTube	Chapters, 31:13
S12	Sanky, 'How I Made an AI Short Drama in under 10 Minutes with ZooClaw + Se	Chapters and description, 7:47
S13	M. Tschochohei and F. Schenker, 'Multimodal Generative AI in the Enterpris	Ch.5 pp.56-75; Ch.7 pp.106-119; Ch.8 pp.127-143
S14	A. Mewada, M. A. Ansari, S. Ahmad and N. Singh (eds), 'AI-Generated Image	Ch.1 pp.1-11; Ch.3 pp.35-44; Ch.7 pp.94-100
S15	Google AI for Developers — Gemini API image generation documentation	https://ai.google.dev/gemini-api/docs/image-generation (
S16	Google AI for Developers — Veo video generation documentation	https://ai.google.dev/gemini-api/docs/veo (verified 6 Se
S17	OpenCV 4.13 Python package and FFmpeg, verified locally on the delivery im	cv2 4.13.0 (opencv-python 4.13.0.92); ffmpeg/ffprobe opt

S6–S12 are creator tutorials, some sponsored: they anchor techniques, not tool choices.

Recommended Follow-On Courses

1

Generative AI for Script Development and Storytelling

WSQ funded · Tertiary Infotech Academy

2

Creating Engaging Videos with Generative AI

WSQ funded · Tertiary Infotech Academy

3

Agentic AI for Video Creation

WSQ funded · Tertiary Infotech Academy

4

Generative AI for Photoshop

WSQ funded · Tertiary Infotech Academy

5

Video Editing with After Effects

WSQ funded · Tertiary Infotech Academy

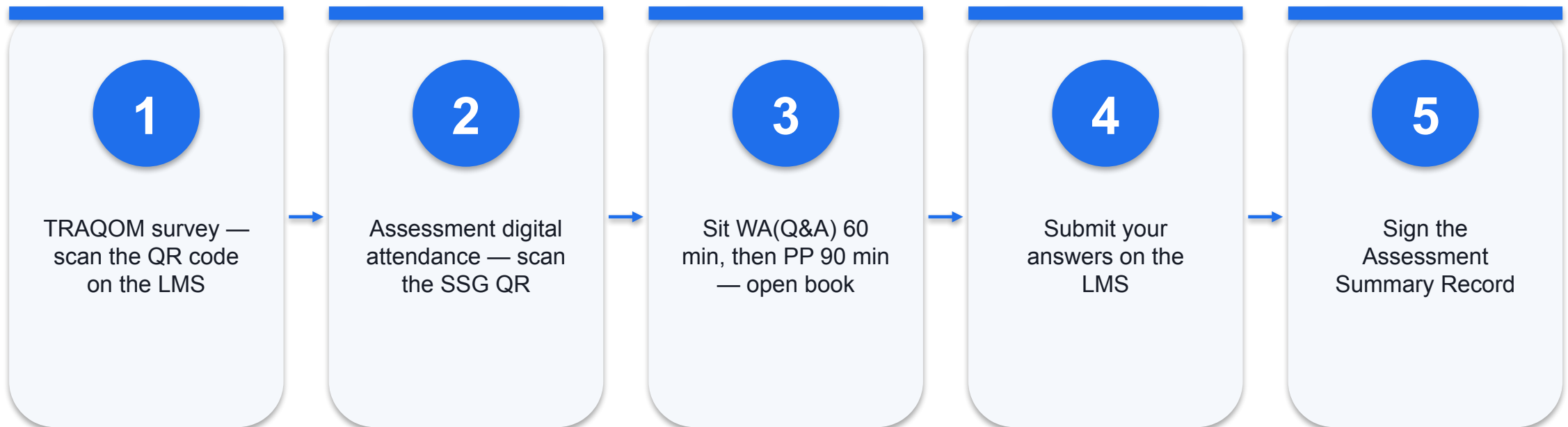
Support

- If you have any enquiries during or after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Telephone: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- Course page: <https://www.tertiarycourses.com.sg/wsq-create-short-video-film-using-ai.html>

Assessment

- Written Assessment WA(Q&A) — 13 open-ended questions covering K1 to K13, 60 minutes.
- Practical Performance (PP) — 5 scenario-based tasks covering A1 to A8, 90 minutes.
- Open book: slides, Learner Guide and approved course materials only.
- Remember to take the Assessment digital attendance before you start.
- Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>
- Sign the Assessment Summary Record when your assessor has completed it.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator displays the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

SEE YOU IN THE NEXT COURSE

Thank You!

You can now plan a short film, generate it reference-first, verify every shot with computer vision, finish it with sound and captions, and defend where it should run.